

模拟电子技术基础

Analog Device

9 Signal Generation

信号的产生

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电子与电气工程系 E³

9.0 Signal generation

Sine-wave/sinusoidal oscillator/generator

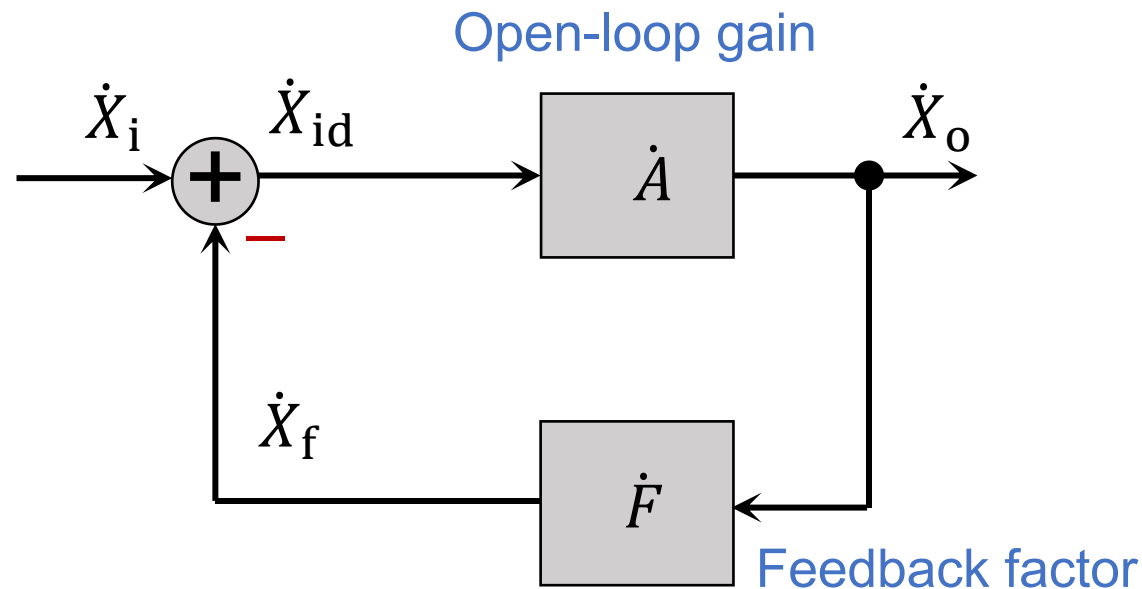
正弦波振荡器

Non-Sine-wave/Non-sinusoidal oscillator/generator

非正弦波振荡器

9.1 Sine-wave oscillator

(1) Self-excited oscillation in negative feedback circuits 负反馈电路中的自激振荡



Close-loop gain

$$\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$
$$\dot{X}_{id} = \dot{X}_i - \dot{X}_f$$

Close-loop gain

$$\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$

$$\dot{X}_{id} = \dot{X}_i - \dot{X}_f$$

At **midband**, negative feedback: $|1 + \dot{A}\dot{F}| > 1$.

$$\left\{ \begin{array}{l} |\dot{X}_{id}| = |\dot{X}_i| - |\dot{X}_f| \\ \varphi_A + \varphi_F = 2n\pi \end{array} \right.$$

At **low or high frequency band**, $\dot{A}$ and $\dot{F}$ are complex numbers, and $|1 + \dot{A}\dot{F}| < 1$ is possible.

Self-excited oscillation 自激振荡: $1 + \dot{A}\dot{F} = 0$

There is output AC signal when no input AC signal.

Self-excited oscillation 自激振荡: $\dot{A}\dot{F} = -1$

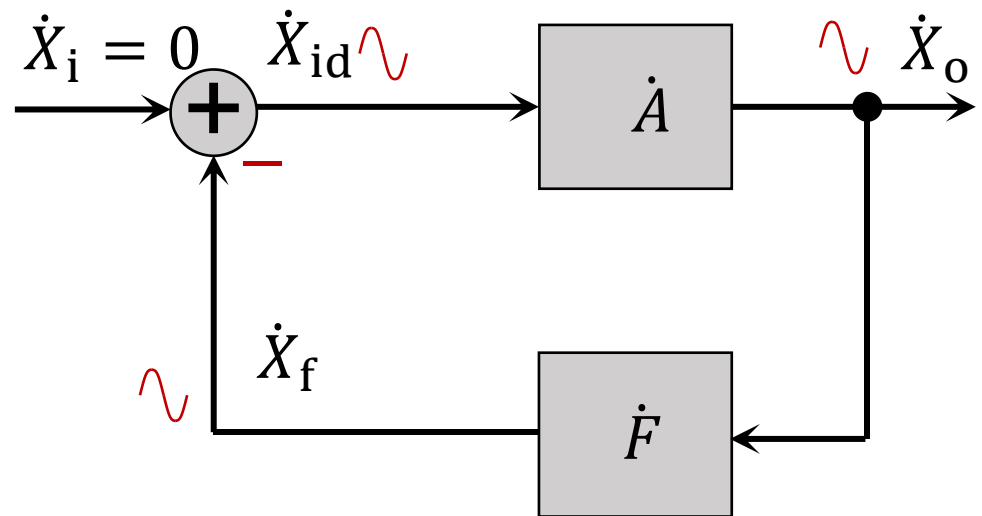
$$\begin{cases} |\dot{A}\dot{F}| = 1 \\ \varphi_A + \varphi_F = 2n\pi + \pi \end{cases}$$

$|\dot{X}_{id}| = |\dot{X}_i| + |\dot{X}_f|$: Negative feedback $\rightarrow$ Positive feedback

When $\dot{A}\dot{F} = -1$

$$\dot{X}_f = \dot{A}\dot{F}\dot{X}_{id} = -\dot{X}_{id}$$

When $\dot{X}_i = 0$,
perturbation at $\dot{X}_o$ cannot
be amplified



Start-oscillation condition 起振条件:

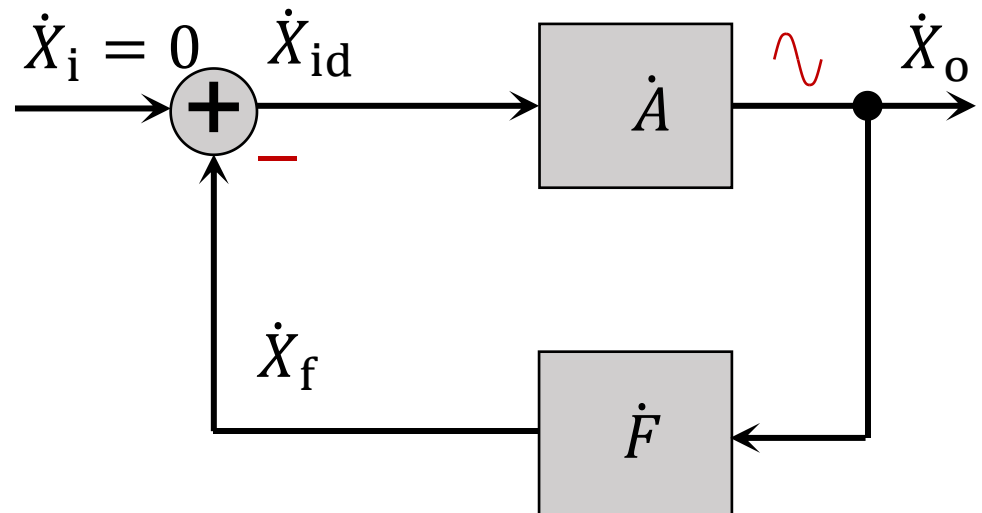
$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| > 1 \quad \text{Perturbation signal can be amplified} \\ \varphi_A + \varphi_F = 2n\pi + \pi \end{array} \right. \quad \begin{array}{l} \text{Positive feedback} \\ |\dot{X}_{id}| = |\dot{X}_i| + |\dot{X}_f| \end{array}$$

When $\dot{X}_i = 0$, perturbation leads to:

$$\sim |\dot{X}_o| \uparrow \Rightarrow |\dot{X}_f|$$



$$\sim |\dot{X}_o| \uparrow\uparrow \leftarrow |\dot{X}_{id}|$$

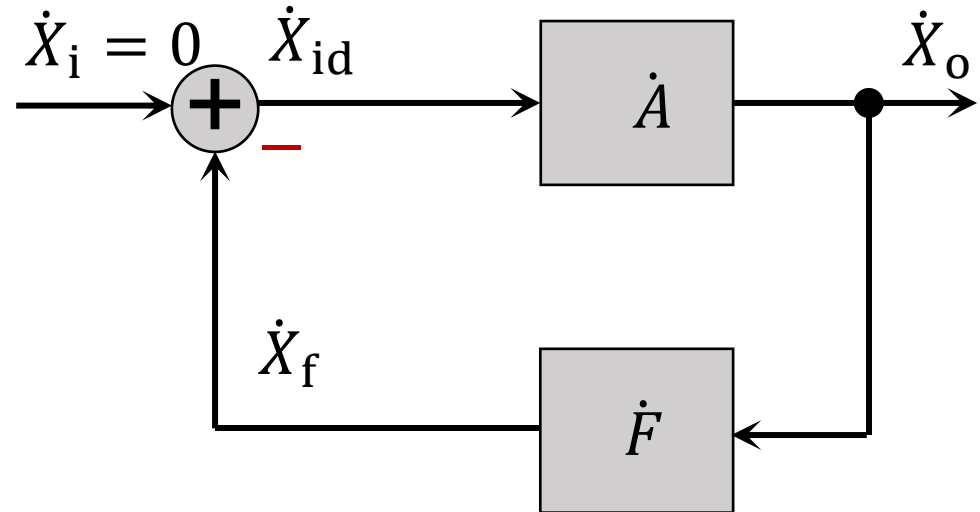


When $\dot{X}_i = 0$, perturbation leads to:

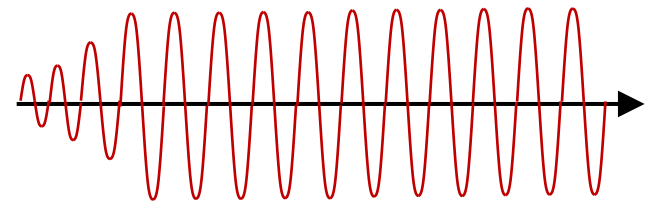
$\sim |\dot{X}_o| \uparrow \Rightarrow |\dot{X}_f|$



$\sim |\dot{X}_o| \uparrow\uparrow \Leftarrow |\dot{X}_{id}|$



When $|\dot{X}_{id}|$ is large enough, amplifier starts to enter saturation region, and $|\dot{A}|$ decrease till $|\dot{A}\dot{F}| = 1$.



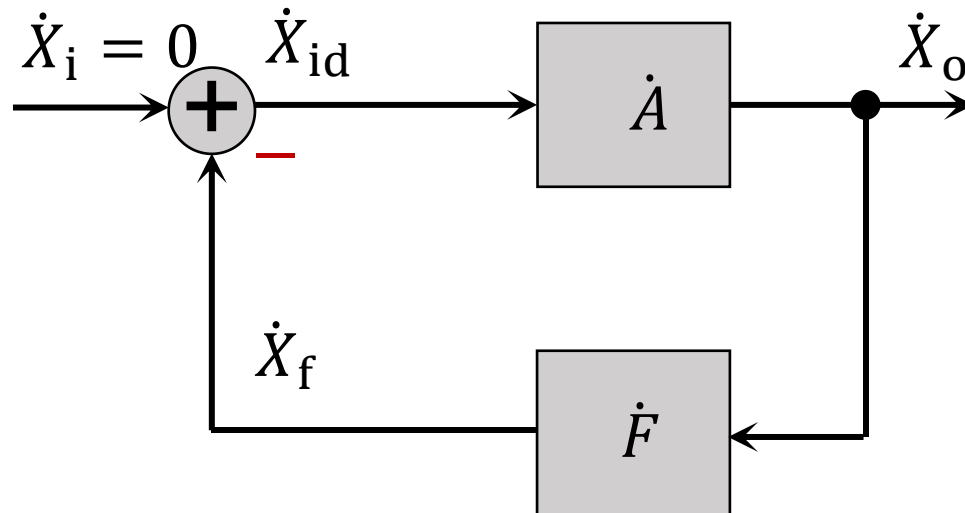
This is the self-excited oscillation

When oscillation is stable:

$$\dot{X}_f = -\dot{A}\dot{F}\dot{X}_{id}$$

$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| = 1 \\ \varphi_A + \varphi_F = 2n\pi + \pi \end{array} \right.$$

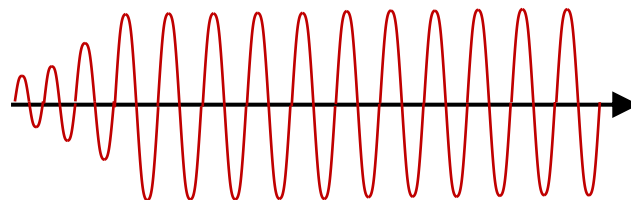
$$\dot{X}_f = \dot{X}_{id}$$



Start-oscillation condition

起振条件:

$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| > 1 \\ \varphi_A + \varphi_F = 2n\pi + \pi \end{array} \right.$$

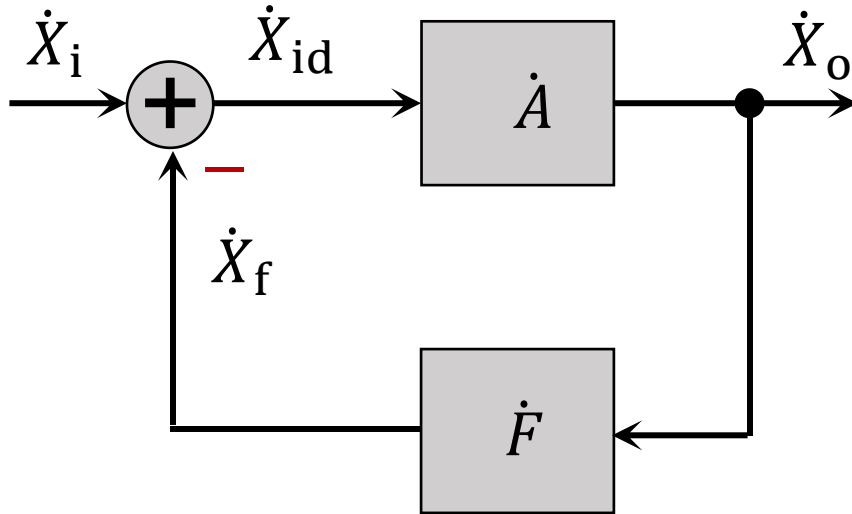


Drawbacks of self-excited oscillation in negative feedback circuits:

Oscillation frequency cannot be well controlled.

Solutions: use **positive** feedback circuits

Negative feedback

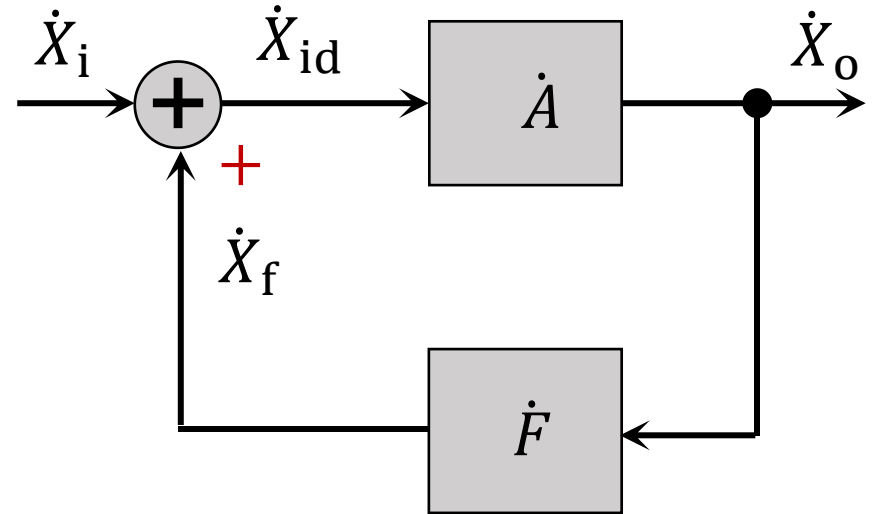


$$\dot{X}_{id} = \dot{X}_i - \dot{X}_f$$

$$\dot{A}_f = \frac{\dot{A}}{1 + \dot{A}\dot{F}}$$

$$|\dot{X}_{id}| = |\dot{X}_i| - |\dot{X}_f|$$

Positive feedback



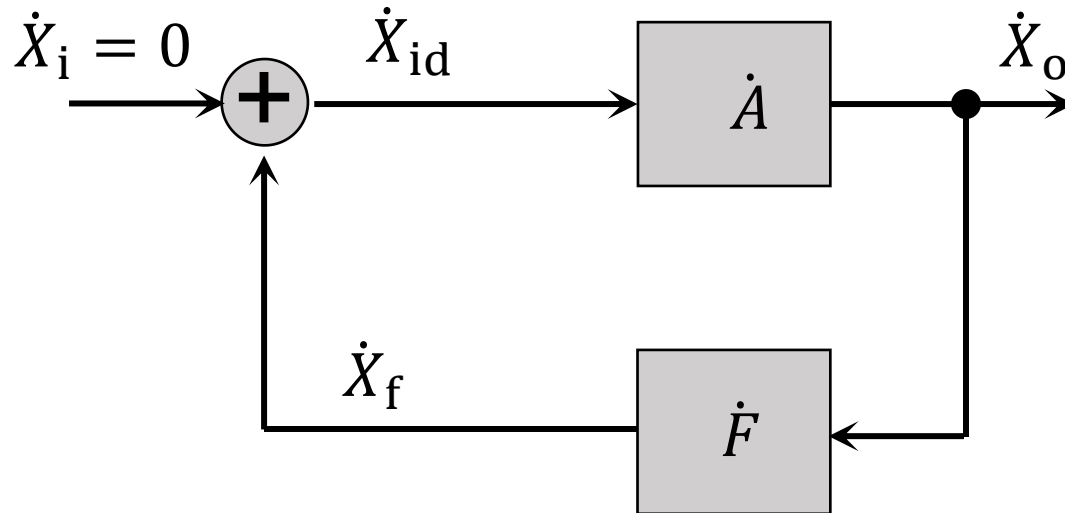
$$\dot{X}_{id} = \dot{X}_i + \dot{X}_f$$

$$\dot{A}_f = \frac{\dot{A}}{1 - \dot{A}\dot{F}}$$

$$|\dot{X}_{id}| = |\dot{X}_i| + |\dot{X}_f|$$

$$\dot{A}\dot{F} > 0$$

(2) Positive feedback circuits 正反馈电路中的自激振荡



$$\dot{X}_{id} = \dot{X}_i + \dot{X}_f$$

$$\dot{A}_f = \frac{\dot{A}}{1 - \dot{A}\dot{F}}$$

$$|\dot{X}_{id}| = |\dot{X}_i| + |\dot{X}_f|$$

When oscillation is stable:

$$\dot{A}\dot{F} = 1$$

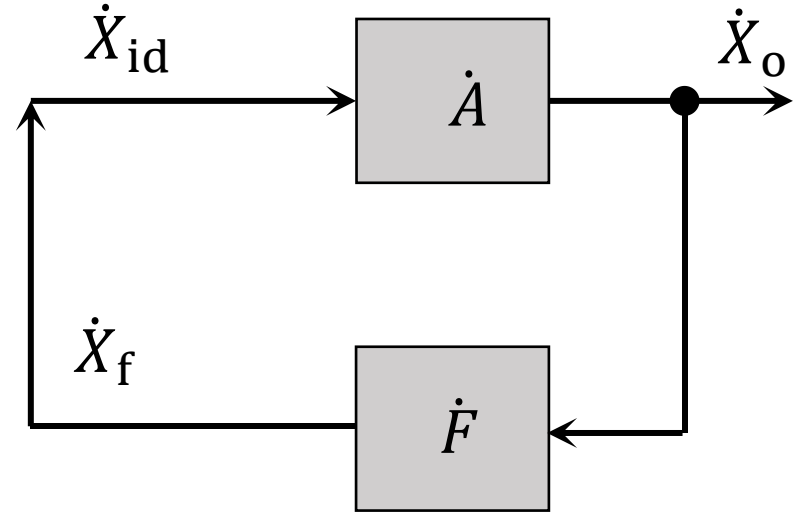
$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| = 1 \\ \varphi_A + \varphi_F = 2n\pi \end{array} \right.$$

When oscillation is stable:

$$\dot{X}_f = \dot{A}\dot{F}\dot{X}_{id}$$

$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| = 1 \\ \varphi_A + \varphi_F = 2n\pi \end{array} \right.$$

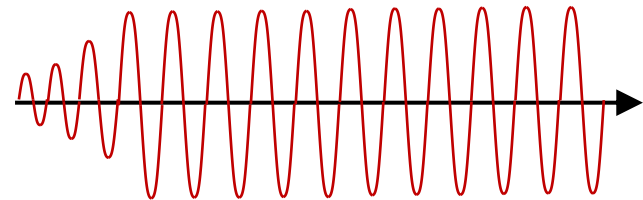
$$\dot{X}_f = \dot{X}_{id}$$



Start-oscillation condition

起振条件:

$$\left\{ \begin{array}{l} |\dot{A}\dot{F}| > 1 \\ \varphi_A + \varphi_F = 2n\pi \end{array} \right.$$



How to control oscillation frequency f_0 ?

$$\text{For } f = f_0, \begin{cases} |\dot{A}\dot{F}| > 1 \\ \varphi_A + \varphi_F = 2n\pi \end{cases}$$

$$\text{For } f \neq f_0, \begin{cases} |\dot{A}\dot{F}| < 1 \\ \varphi_A + \varphi_F \neq 2n\pi \end{cases}$$

Frequency selection network 选频网络

Band pass filter 带通滤波

Sine-wave oscillator contains:

Amplifier circuit 放大电路

Positive feedback network 正反馈网络

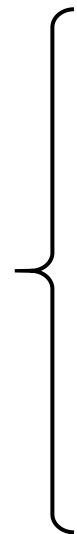
Frequency selection network 选频网络

Amplitude stabilization link 稳幅环节

Can be
combined!
合二为一

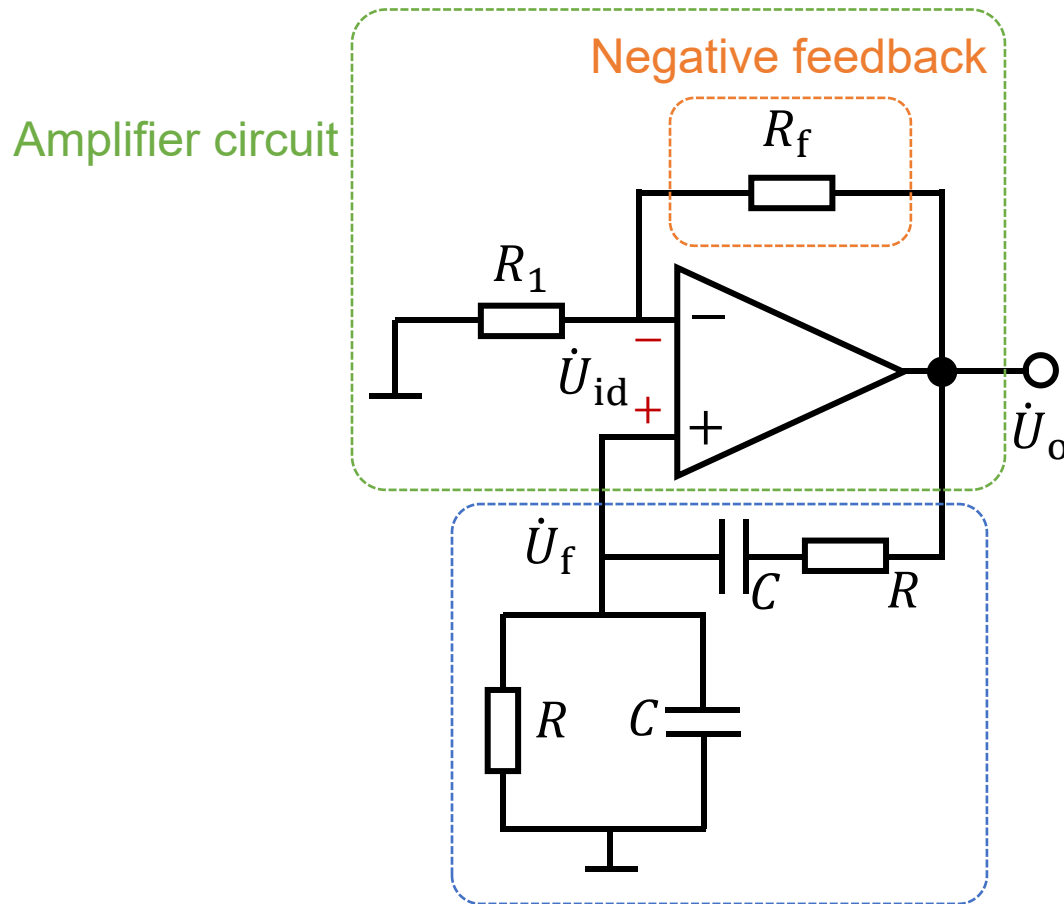
Use the non-linear characteristic of electronic devices

Sine-wave oscillator types:

- 
- RC sine-wave oscillator
 - LC sine-wave oscillator
 - Crystal oscillator 晶体振荡器

9.2 RC sine-wave oscillator

Wien bridge oscillator 文氏电桥振荡器

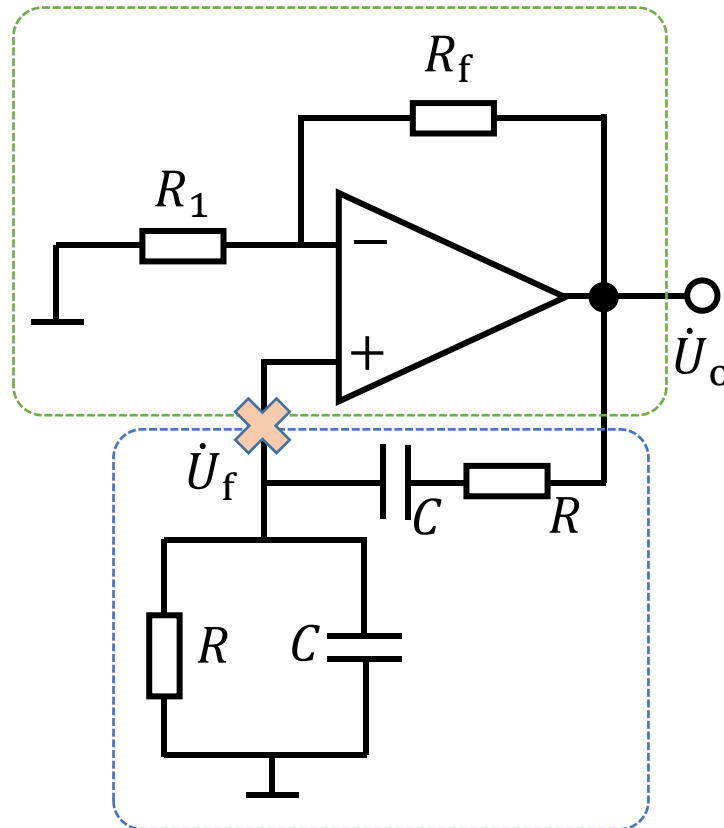


Positive feedback + Frequency selection

Resonant frequency? $\dot{A} = ?$ $\dot{F} = ?$

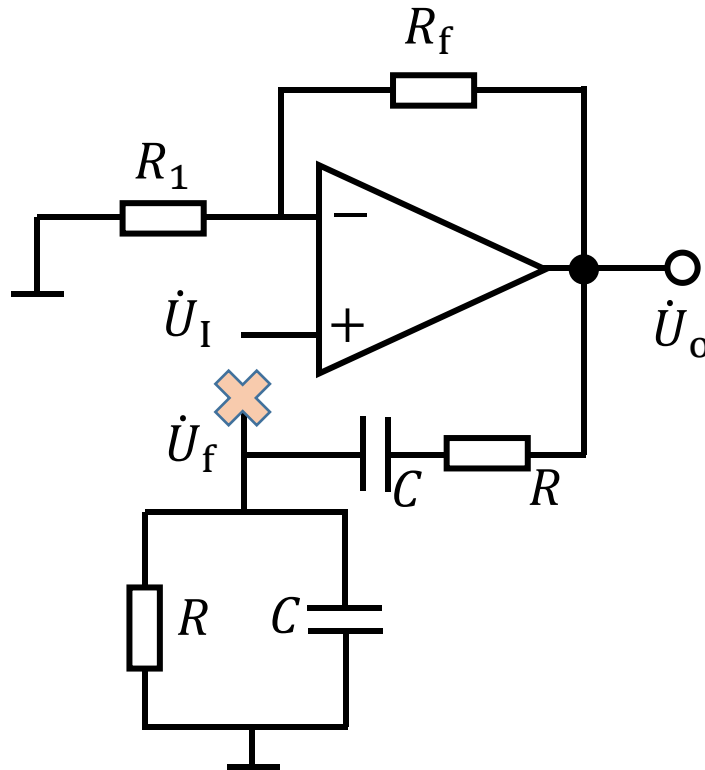
Separation method 分离法

Amplifier circuit



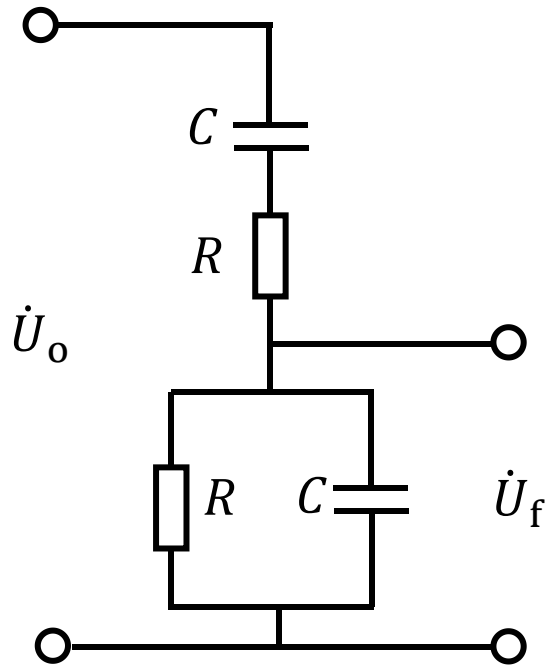
Positive feedback + Frequency selection

Amplifier circuit



$$\left\{ \begin{array}{l} \dot{A} = \frac{\dot{U}_O}{\dot{U}_I} = 1 + \frac{R_f}{R_1} \\ \varphi_A = 0 \end{array} \right.$$

Frequency selection network



$$\dot{F} = \frac{\dot{U}_f}{\dot{U}_o}$$

$$= \frac{R // \frac{1}{j\omega C}}{R // \frac{1}{j\omega C} + R + \frac{1}{j\omega C}}$$

$$= \frac{1}{3 + j(\omega RC - \frac{1}{\omega RC})}$$

$$= \frac{1}{3 + j(\frac{f}{f_0} - \frac{f_0}{f})}, \quad \omega_0 = 2\pi f_0 = \frac{1}{RC}$$

Frequency selection network

$$\dot{F} = \frac{1}{3 + j(\frac{f}{f_0} - \frac{f_0}{f})}, \quad f_0 = \frac{1}{2\pi RC}$$

Amplitude-frequency characteristic

$$|\dot{F}| = \frac{1}{\sqrt{3^2 + (\frac{f}{f_0} - \frac{f_0}{f})^2}}$$

Phase-frequency characteristic

$$\varphi_F = -\arctan \frac{1}{3} \left(\frac{f}{f_0} - \frac{f_0}{f} \right)$$

Frequency selection network

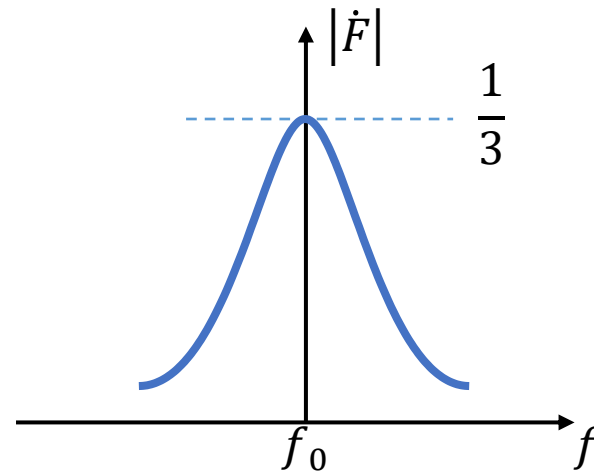
Amplitude-frequency characteristic

$$|\dot{F}| = \frac{1}{\sqrt{3^2 + \left(\frac{f}{f_0} - \frac{f_0}{f}\right)^2}}$$

(a) When $f \rightarrow 0$, $|\dot{F}| \rightarrow 0$

(b) When $f \rightarrow \infty$, $|\dot{F}| \rightarrow 0$

(c) When $f = f_0$, $|\dot{F}| = \frac{1}{3}$



Frequency selection network

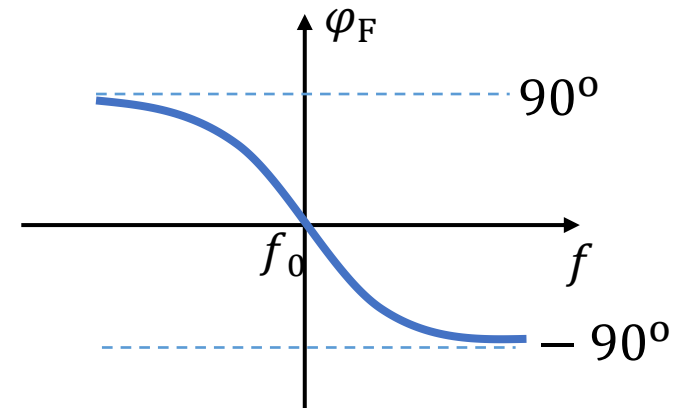
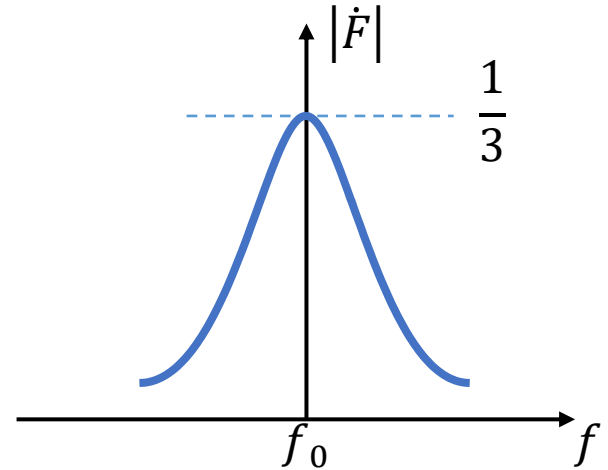
Phase-frequency characteristic

$$\varphi_F = -\arctan \frac{1}{3} \left(\frac{f}{f_0} - \frac{f_0}{f} \right)$$

(a) When $f \rightarrow 0$, $\varphi_F \rightarrow 90^\circ$

(b) When $f \rightarrow \infty$, $\varphi_F \rightarrow -90^\circ$

(c) When $f = f_0$, $\varphi_F = 0$



f_0 : resonant frequency 谐振频率

Working principle:

(a) When $f = f_0$, $\varphi_A = 0$, $\varphi_F = 0$.



$$\varphi = \varphi_A + \varphi_F = 0$$

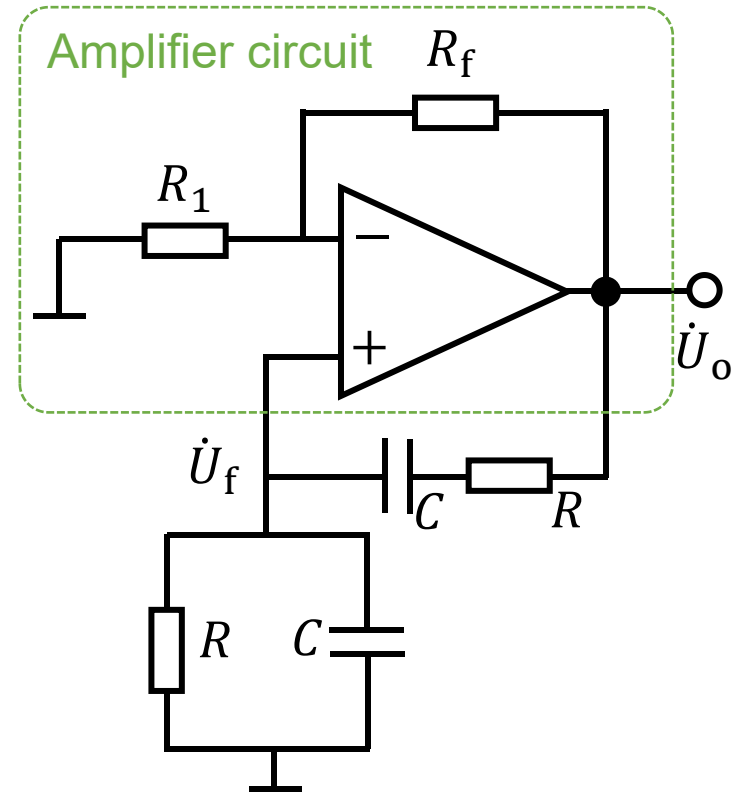
$$\begin{cases} \dot{A} = 1 + \frac{R_f}{R_1} \\ \dot{F} = \frac{1}{3} \end{cases}$$

$\dot{A}\dot{F} > 1$: **Oscillation starts!**



$$\dot{A} = 1 + \frac{R_f}{R_1} > 3$$

(b) Oscillation frequency: $f_0 = \frac{1}{2\pi RC}$



Working principle:

(c) When $f \neq f_0$,

Phase $\varphi_F \neq 2n\pi$.

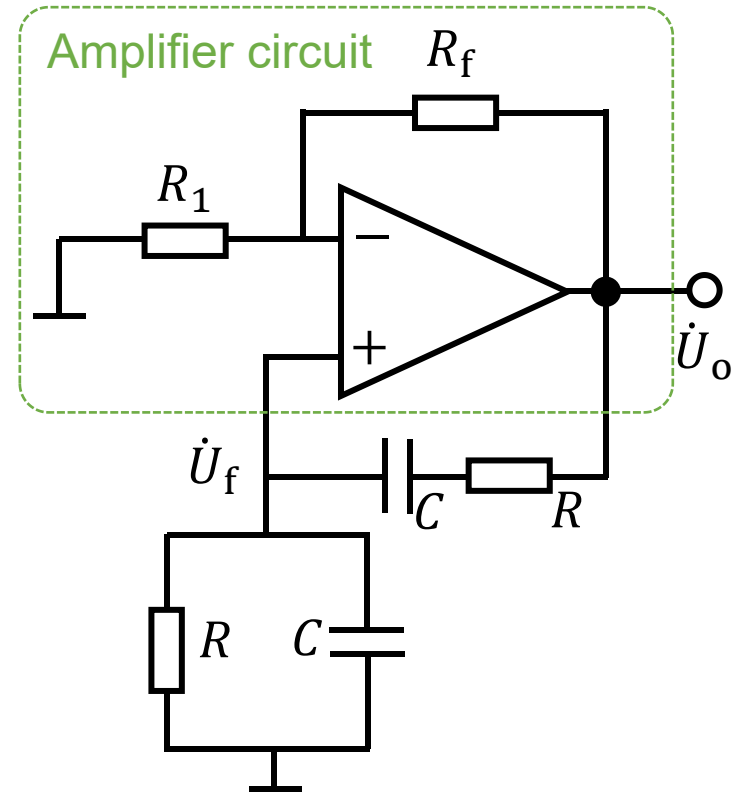
$$\varphi = \varphi_A + \varphi_F \neq 2n\pi$$

$|\dot{F}|$ decrease.



$|\dot{A}\dot{F}|$ decrease

No self-excited oscillation!

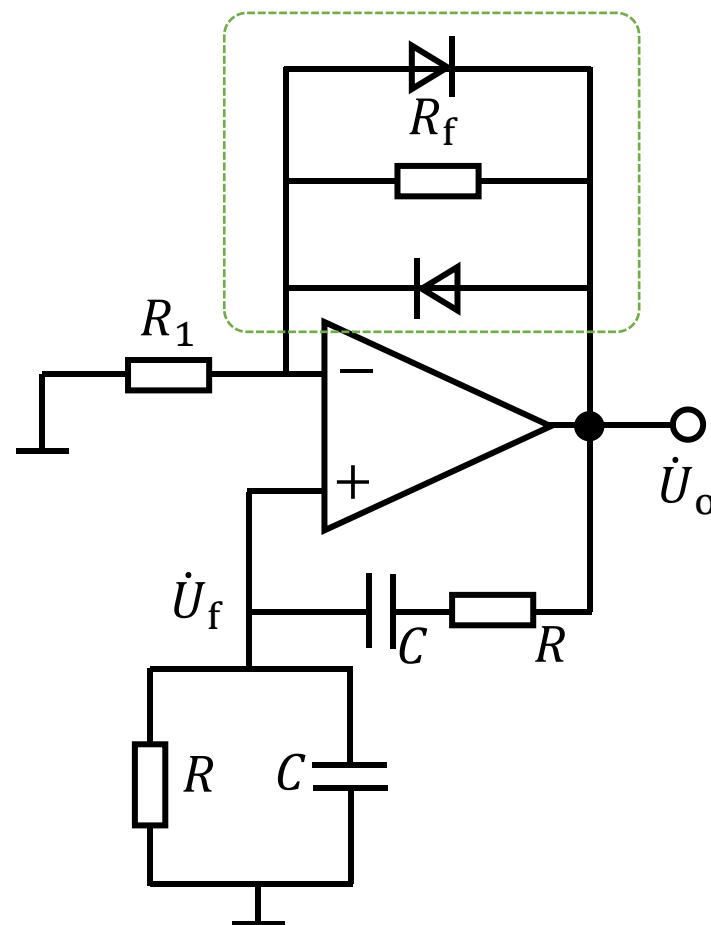
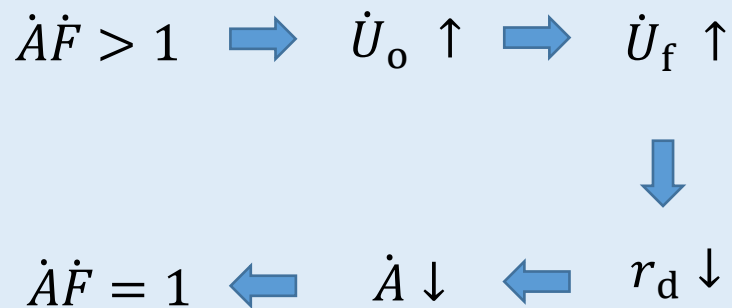


Amplitude stabilization 稳幅措施

Make R_f non-linear! $\dot{A} = 1 + \frac{R_f}{R_1}$

(1) Use two diodes!

$$\dot{A} = 1 + \frac{R_f // r_d}{R_1}$$



Amplitude stabilization 稳幅措施

Make R_f non-linear! $\dot{A} = 1 + \frac{R_f}{R_1}$

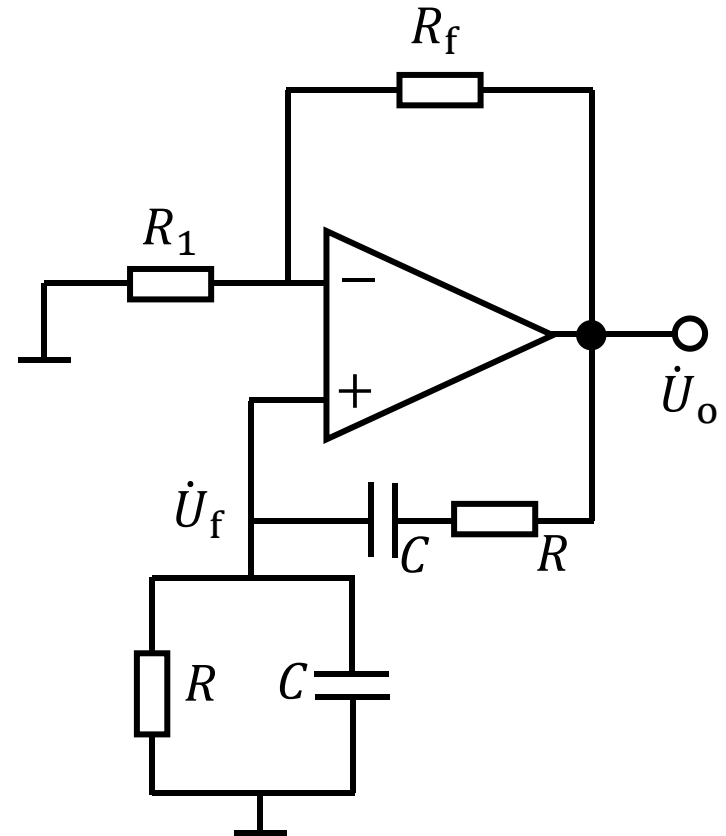
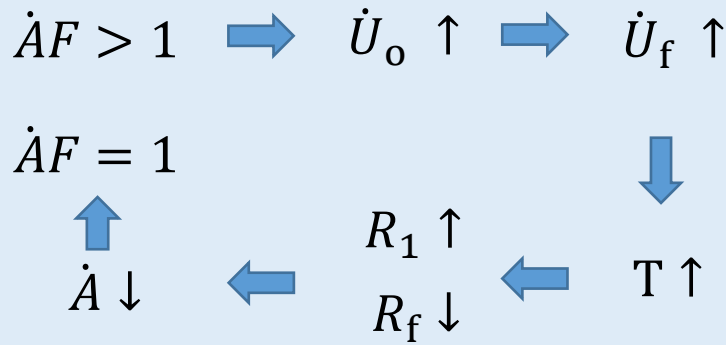
(2) Use thermistor 热敏电阻!

↓

$$\dot{A} = 1 + \frac{R_f}{R_1}$$

Negative TC
负温度系数

Positive TC
正温度系数

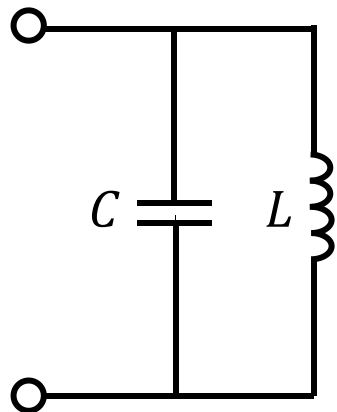


9.3 LC sine-wave oscillator

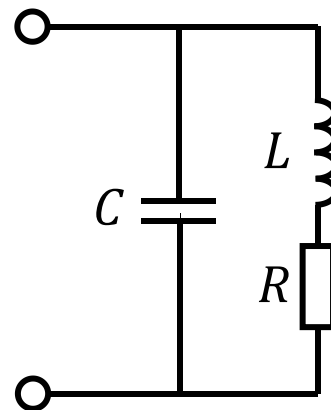
Use inductor and capacitor as frequency selection network

(1) Inductor and capacitor in parallel connection

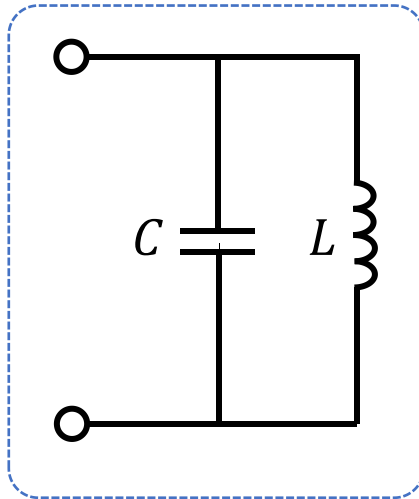
Ideal case



Real case



Ideal case



Variable resistor

Impedance 阻抗 $Z = j\omega L // \frac{1}{j\omega C}$

Admittance 导纳 $Y = \frac{1}{Z} = j\omega C + \frac{1}{j\omega L}$

Resonant frequency: Imaginary part of Y is zero 共振频率: 导纳虚部为0。

$$Y = j\left(\omega C - \frac{1}{\omega L}\right) = 0$$

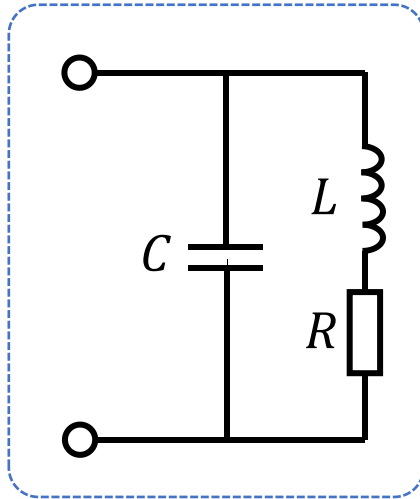


$$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$



$$\mathbf{Z} \rightarrow \infty$$

Real case



Impedance 阻抗

$$Z = (j\omega L + R) // \frac{1}{j\omega C}$$

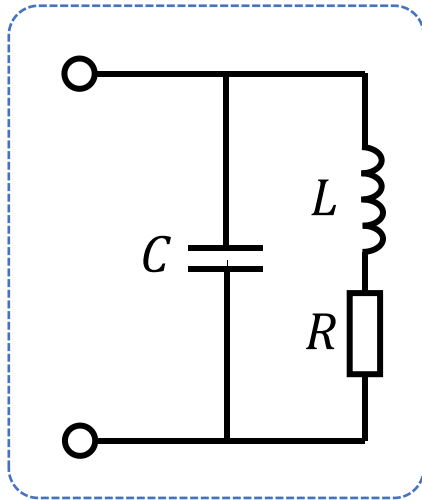
$$Y = \frac{1}{Z} = j\omega C + \frac{1}{j\omega L + R}$$

$$= \frac{R}{R^2 + (\omega L)^2} + j\left[\omega C - \frac{\omega L}{R^2 + (\omega L)^2}\right]$$

Resonant frequency: Imaginary part of Y is zero, Z act as a pure resistor, $\varphi_Z = 0$

$$\omega_0 = \frac{1}{L} \sqrt{\frac{L}{C} - R^2} = \frac{1}{\sqrt{1 + \left(\frac{R}{\omega_0 L}\right)^2}} \frac{1}{\sqrt{LC}}$$

Real case



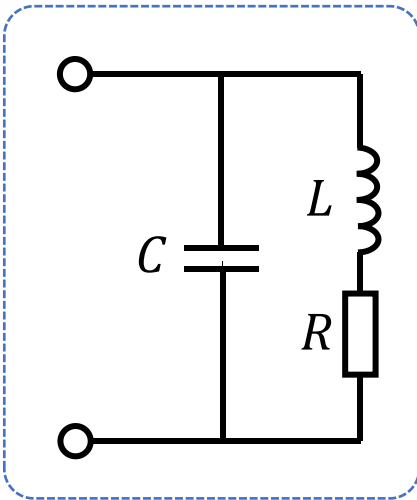
$$\omega_0 = \frac{1}{\sqrt{1 + \left(\frac{R}{\omega_0 L}\right)^2}} \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 + \left(\frac{1}{Q}\right)^2}} \frac{1}{\sqrt{LC}}$$

$$Q = \frac{\omega_0 L}{R} : \text{the quality factor}$$

$$\text{When } Q \gg 1, \quad \omega_0 = 2\pi f_0 \approx \frac{1}{\sqrt{LC}}$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Real case



$$\text{When } f = f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Impedance :

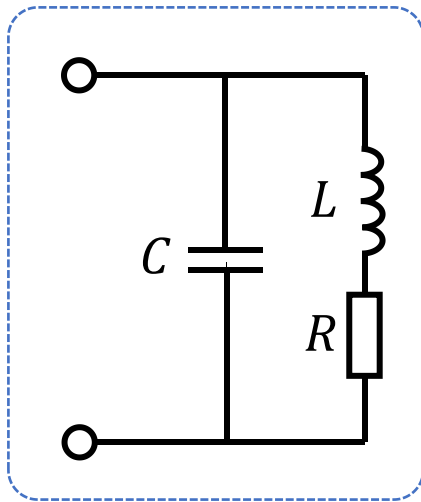
$$Z_0 = \frac{1}{Y_0} = \frac{R^2 + (\omega_0 L)^2}{R}$$
$$= R + Q^2 R$$

Act as a pure resistor 等效为纯电阻

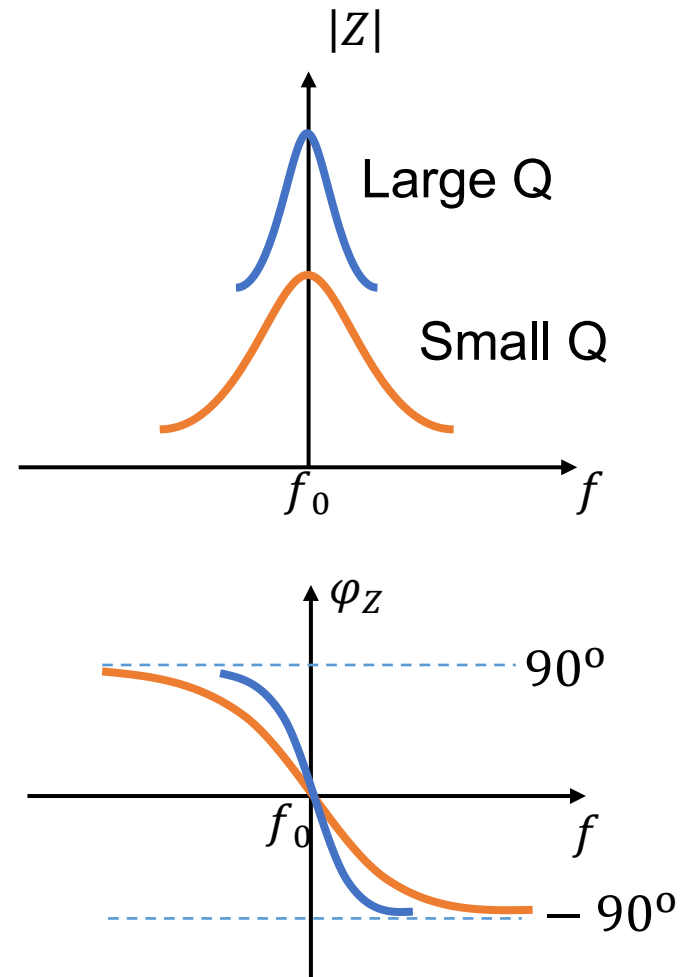
$$\text{When } Q \gg 1, \quad Z_0 \approx Q^2 R$$

$$Z_0 \approx QX_C \approx QX_L$$

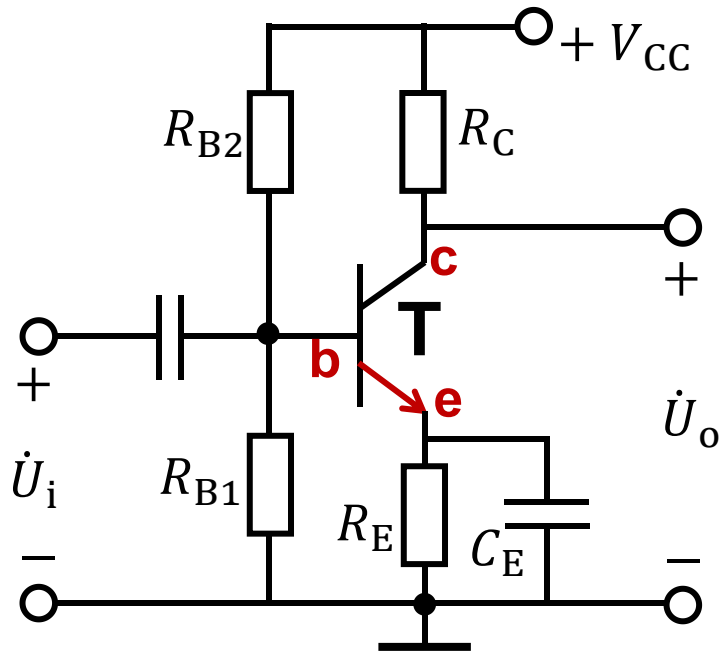
Real case



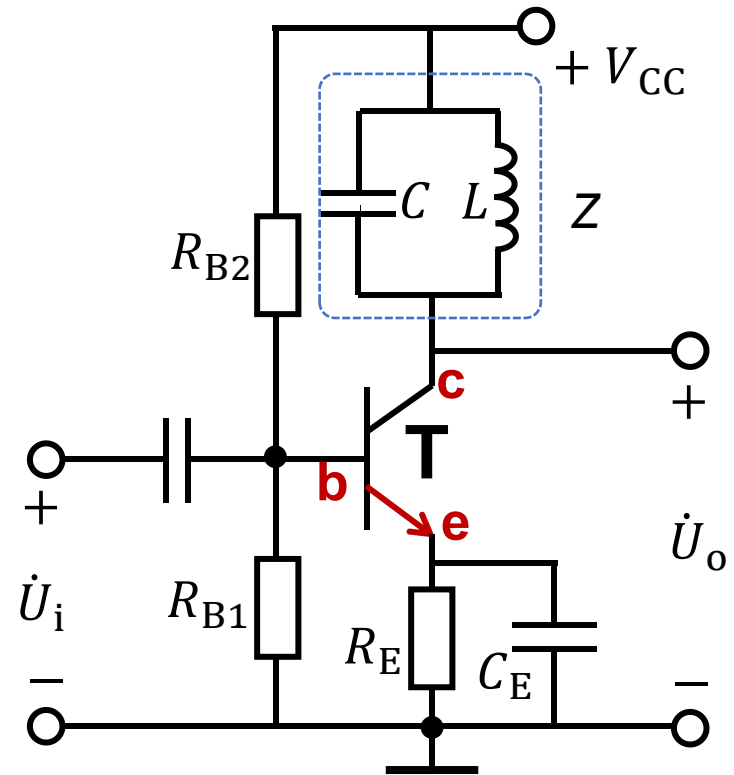
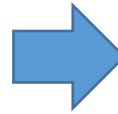
Larger Q, better selectivity!



Amplifier circuit with LC frequency selection network

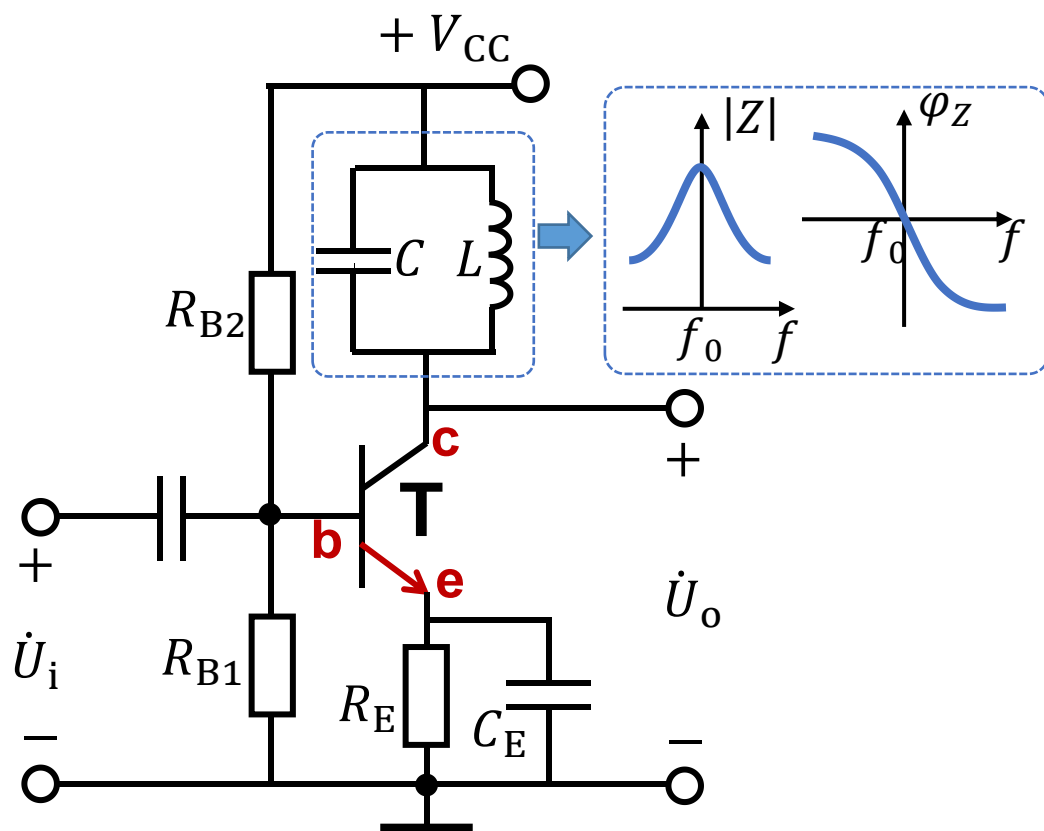


$$\dot{A}_u = - \frac{\beta(R_L || R_C)}{r_{be}}$$



$$\dot{A}_u = - \frac{\beta(R_L || Z)}{r_{be}}$$

Amplifier circuit with LC frequency selection network



$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = - \frac{\beta(R_L \parallel Z)}{r_{be}}$$

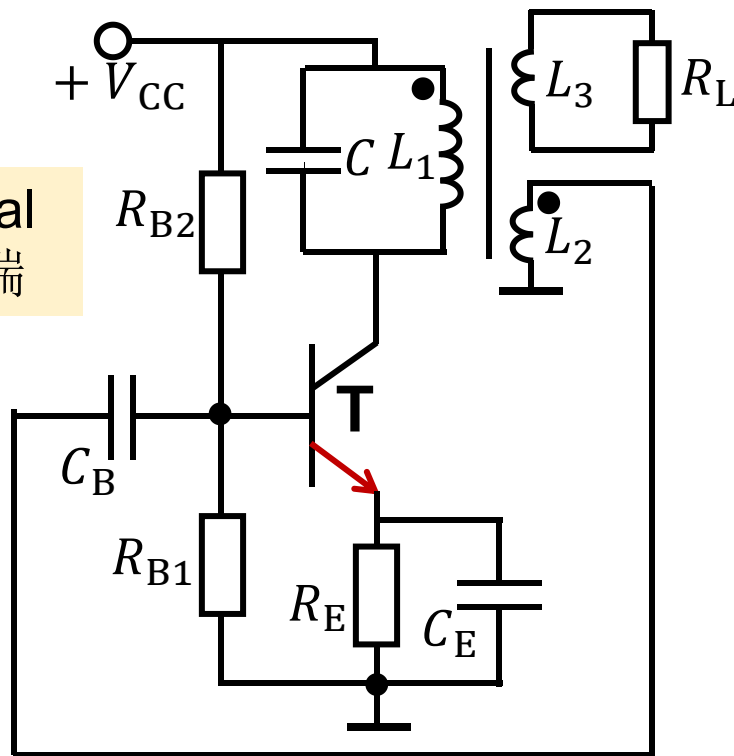
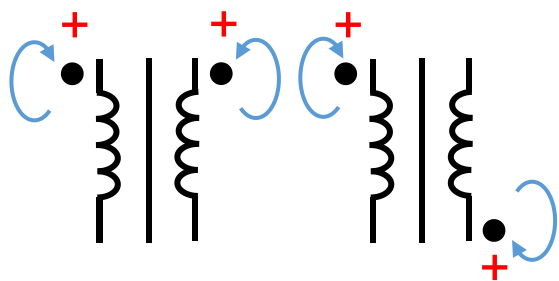
Larger Z , larger $\dot{A}_u$

$$\text{When } f = f_0 = \frac{1}{2\pi\sqrt{LC}},$$

$$\left\{ \begin{array}{l} \dot{A}_u \text{ is maximum} \\ \text{Phase of } \dot{A}_u = 180^\circ \end{array} \right.$$

(2) Transformer feedback LC oscillator 变压器反馈式LC振荡电路

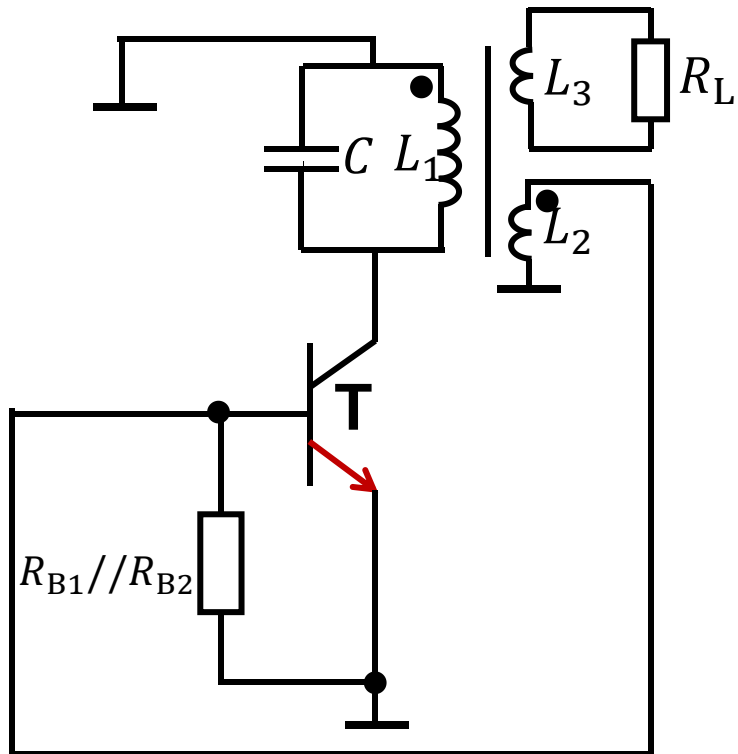
Dotted terminal of mutual inductor 互感线圈同名端



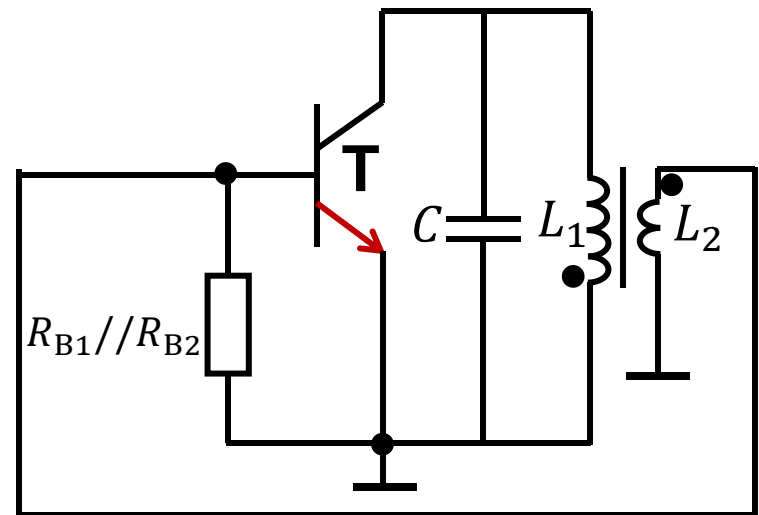
Feedback network?

Positive or negative for AC circuit?

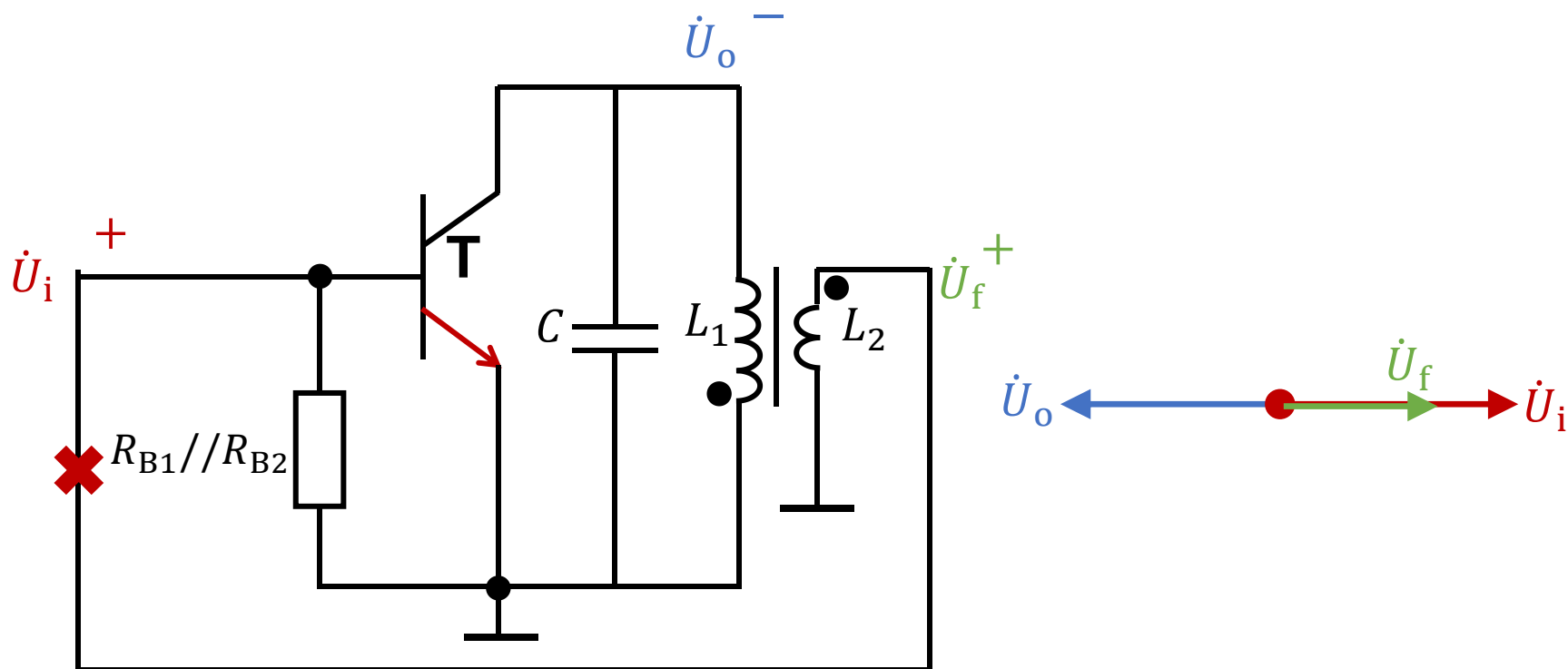
AC circuit



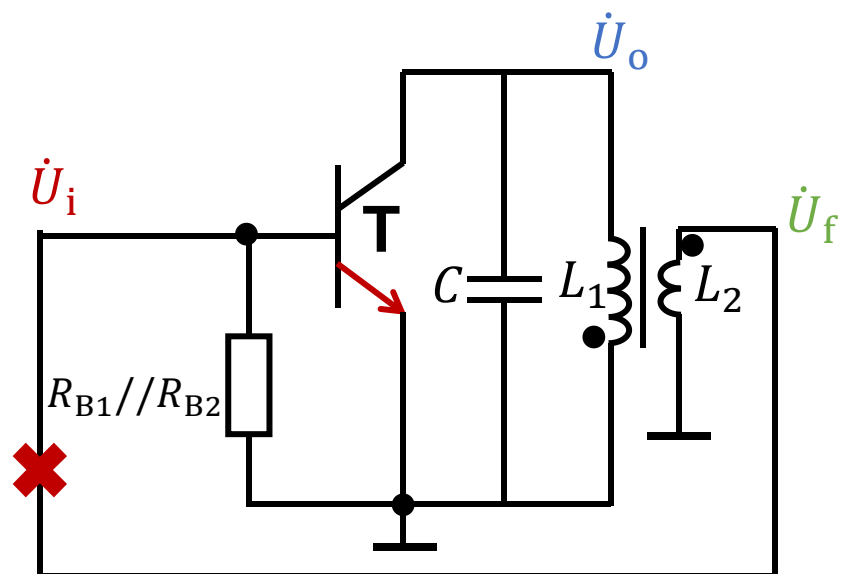
Ignore load resistor R_L



Instantaneous polarity method 瞬时极性法



Ignore mutual inductance
between L1 and L2 忽略互感



Start-oscillation condition: 起振条件

(a) $\dot{A}\dot{F} > 1$

$$\left\{ \begin{array}{l} \dot{A} = \frac{\dot{U}_o}{\dot{U}_i} = -\frac{\beta Z}{r_{be}} \\ \dot{F} = \frac{\dot{U}_f}{\dot{U}_o} = -\frac{L_2}{L_1} \end{array} \right.$$

(b) $\varphi = 2n\pi$

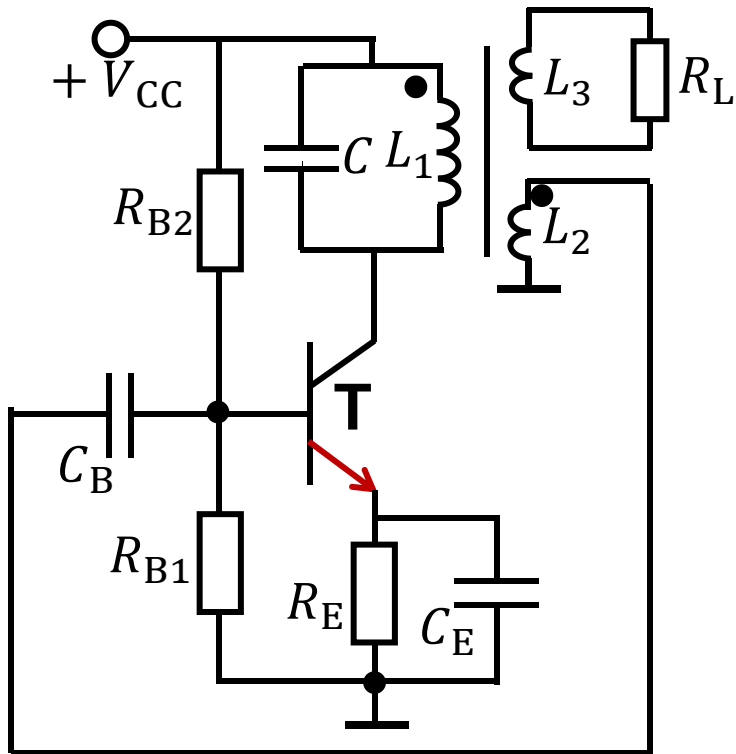


Phase of Z is 0



Oscillation frequency:

$$f = f_0 = \frac{1}{2\pi\sqrt{LC}}$$



Merits:

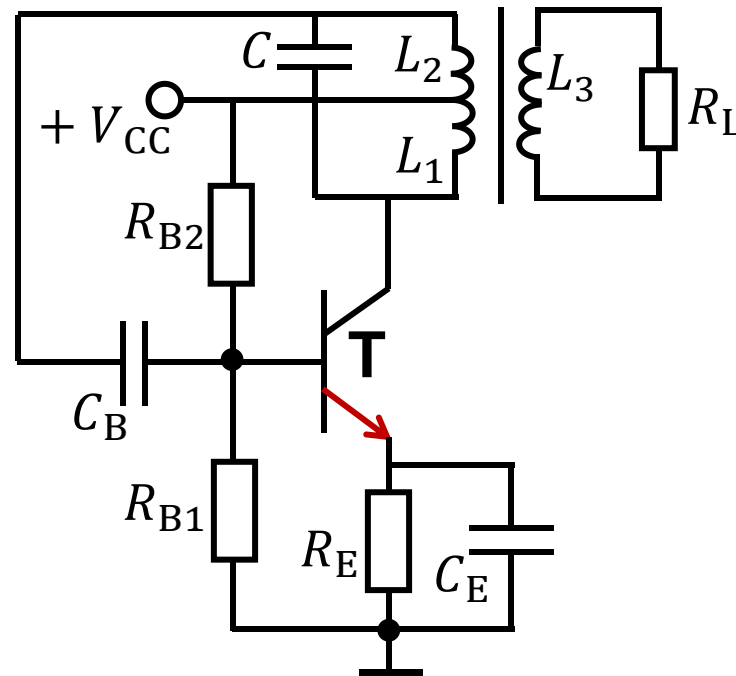
- Liability to generate oscillation
- High oscillation frequency
- High quality of output waveform

Drawback:

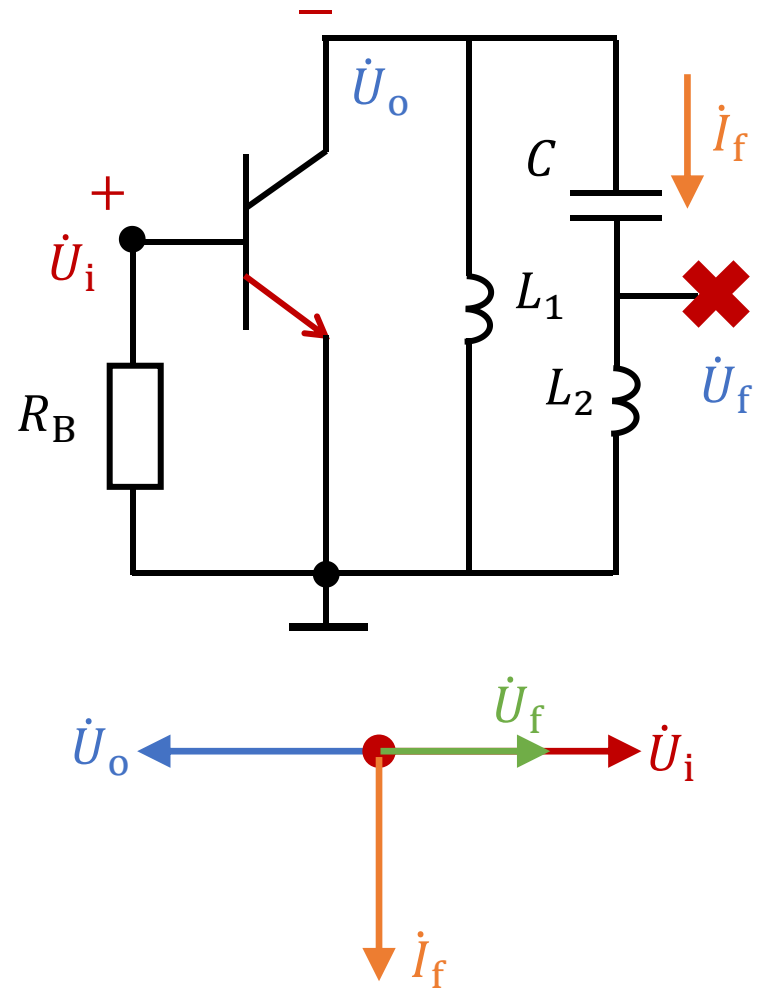
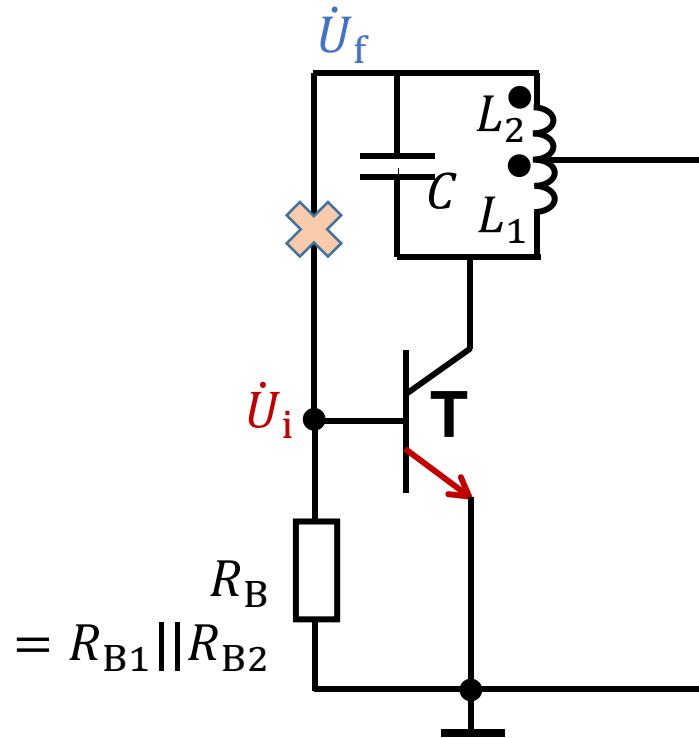
Power loss is high

(3) Hartley feedback oscillator 电感反馈式/三点式振荡电路

Positive or negative for AC circuit?

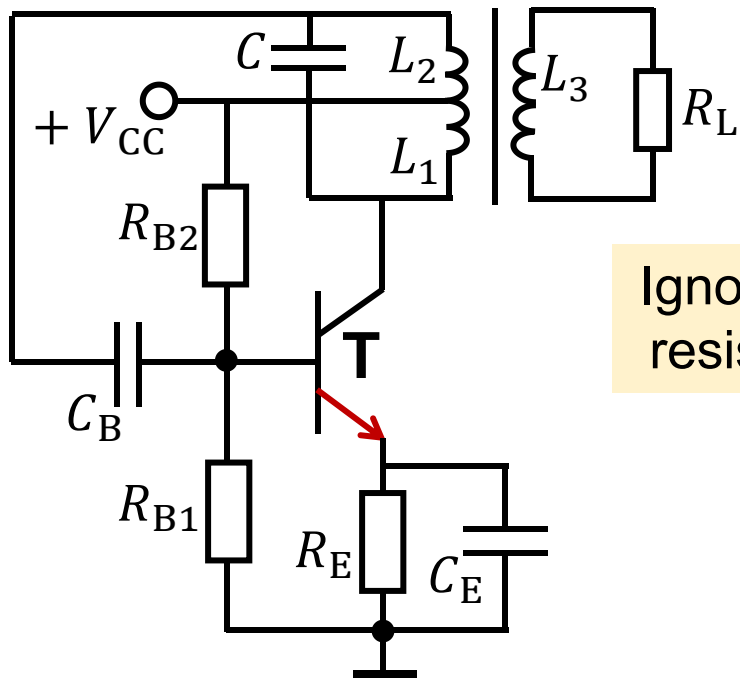


AC circuit

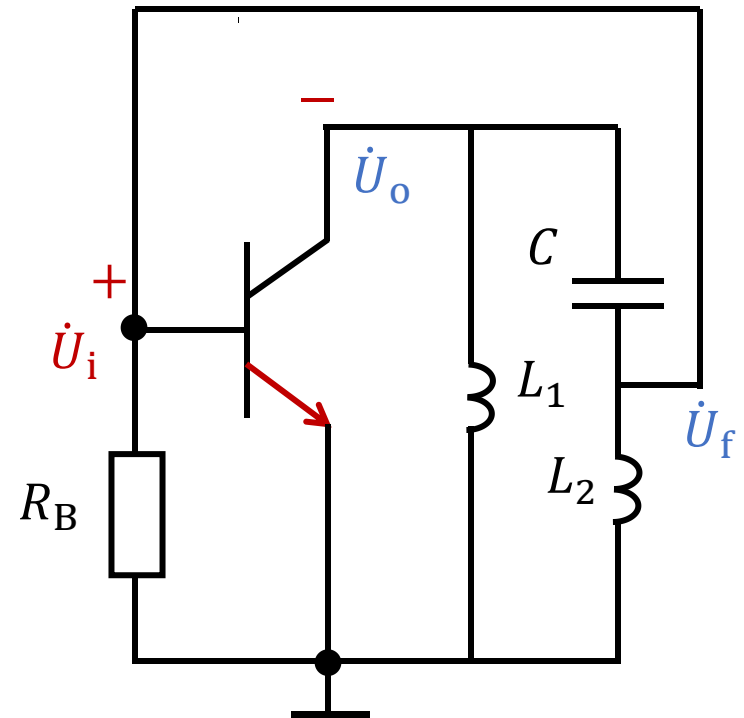


$\dot{U}_f$ and $\dot{U}_i$ have the same phase:
positive!

Resonant frequency? $\dot{A} = ?$ $\dot{F} = ?$

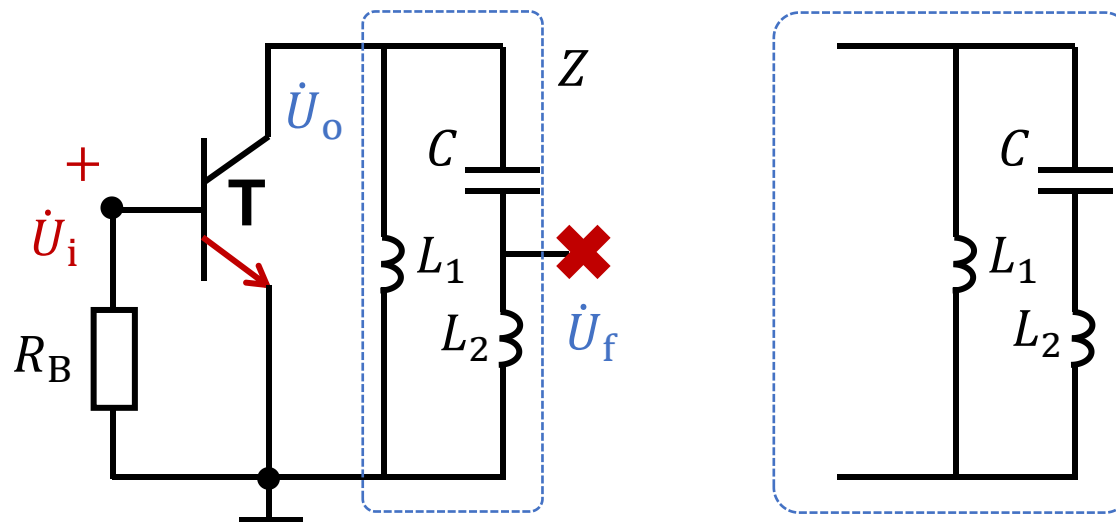


Ignore load resistor R_L



Inductor: $R \rightarrow 0$

Ignore mutual inductance between L_1 and L_2
为了简略忽略互感（实际情况是一定要考虑互感）



Voltage gain: $\dot{A} = -\frac{\beta Z}{r_{be}}$

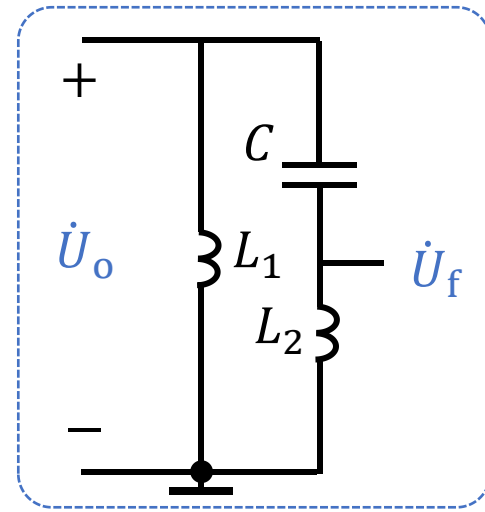
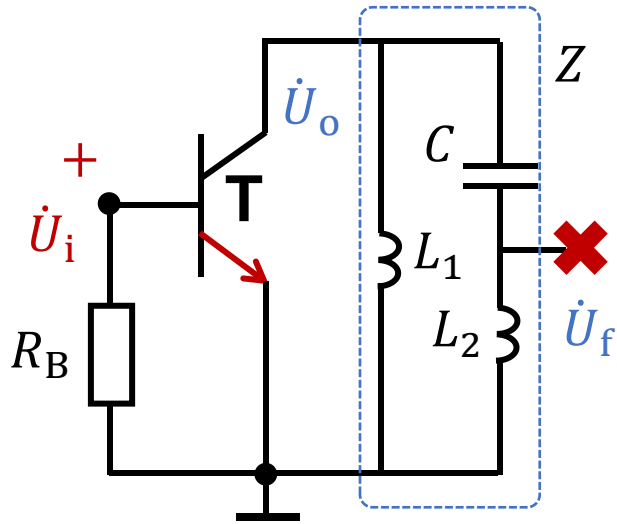
Resonant frequency: $f_0 = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$

$Z = (j\omega L_2 + \frac{1}{j\omega C}) // j\omega L_1$

$Y = \frac{1}{Z} = j(\frac{1}{\omega C} - \omega L_2 - \frac{1}{\omega L_1})$

$\begin{cases} Z \rightarrow \infty \\ Y \rightarrow 0 \end{cases}$

Positive feedback network:



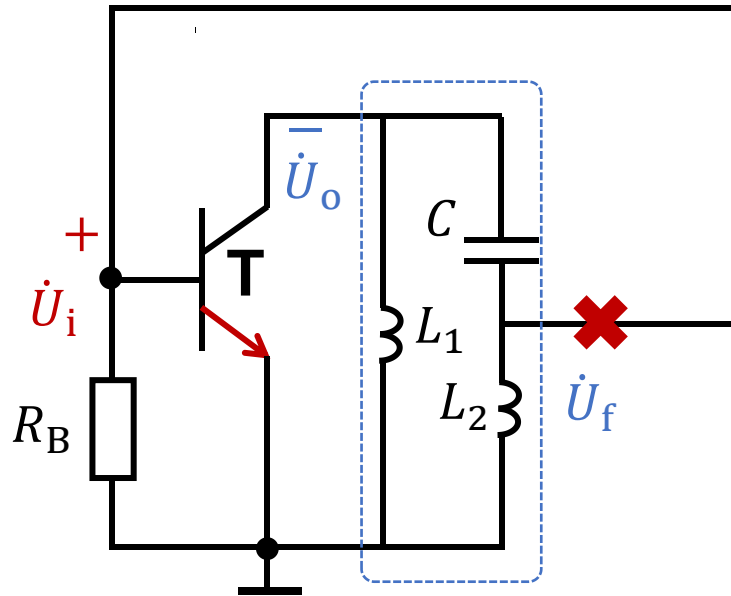
$$\dot{F} = -\frac{L_2}{L_1}$$

$$f_0 = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

$$\dot{F} = \frac{\dot{U}_f}{\dot{U}_o} = \frac{j\omega L_2}{\frac{1}{j\omega C} + j\omega L_2} = \frac{1}{1 - \frac{1}{\omega^2 L_2 C}}$$

Start-oscillation condition 起振条件:

(a) $\dot{A}\dot{F} > 1$



$$\begin{cases} \dot{A} = -\frac{\beta Z}{r_{be}} \\ \dot{F} = -\frac{L_2}{L_1} \end{cases}$$

(b) $\varphi = 2n\pi$

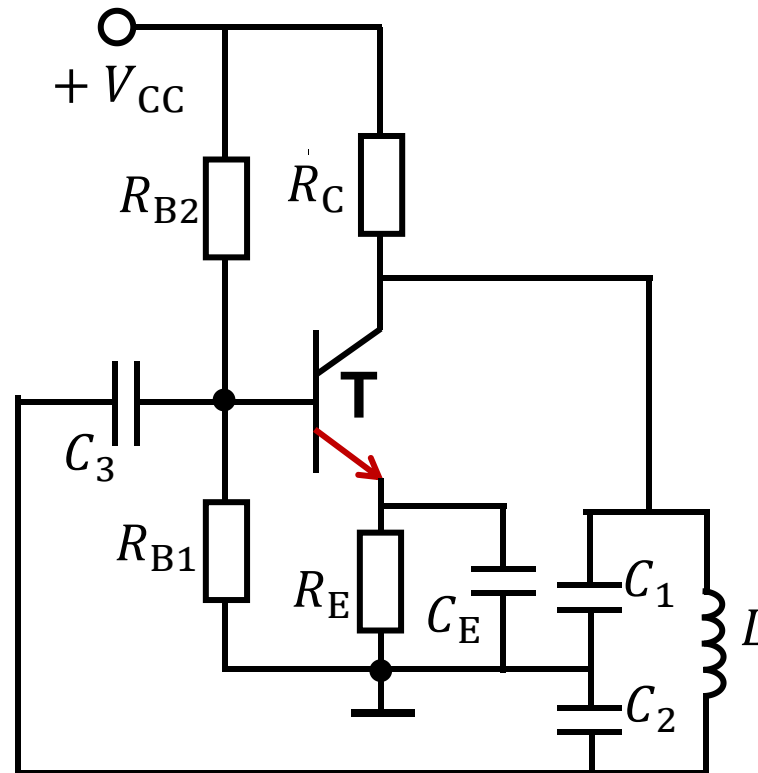
At resonant frequency: $f_0 \approx \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$

$$\varphi_A = \pi, \quad \varphi_F = \pi.$$

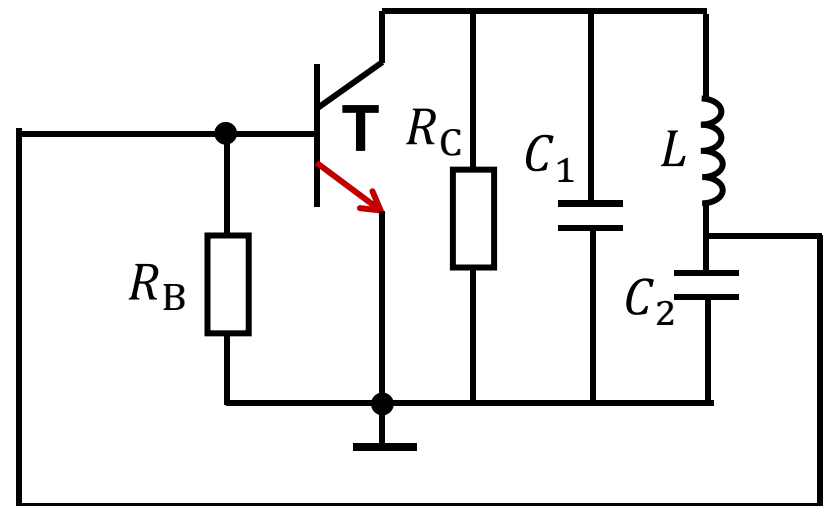
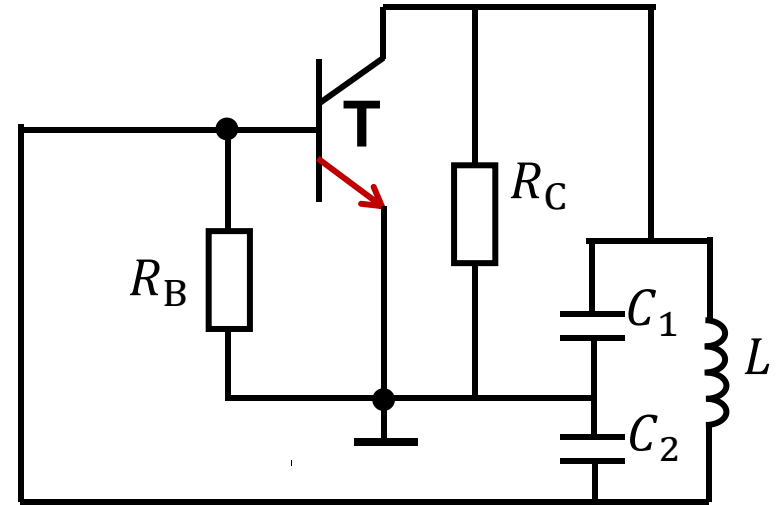
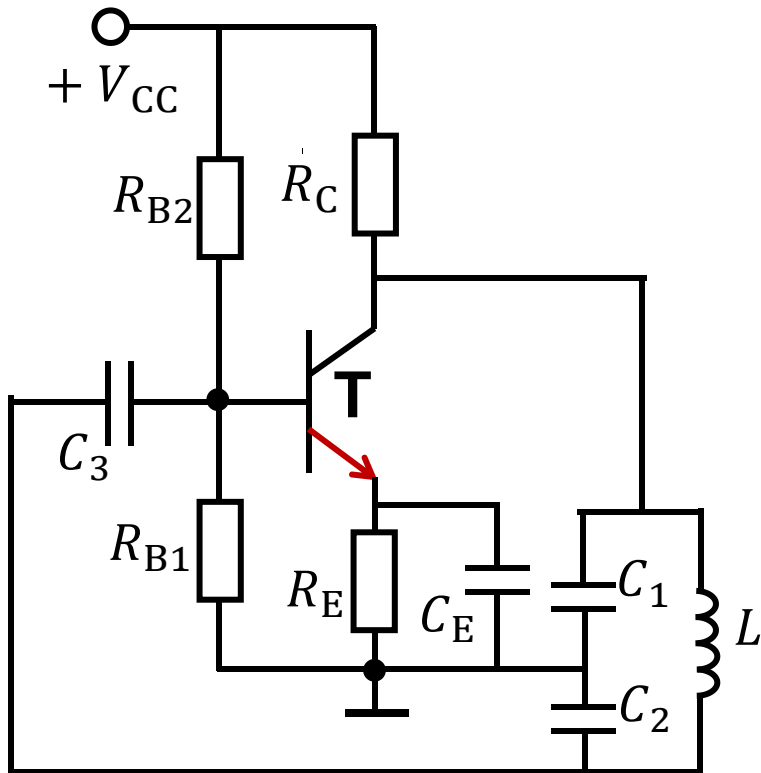
$$\varphi = \varphi_A + \varphi_F = 2\pi$$

(4) Colpitts feedback oscillator 电容反馈式/三点式振荡电路

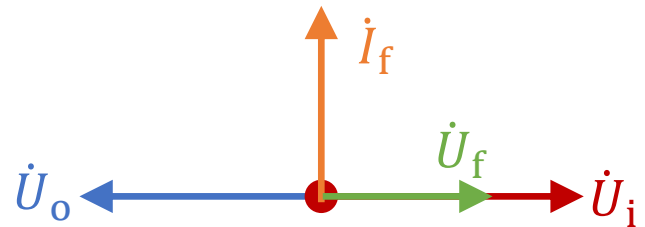
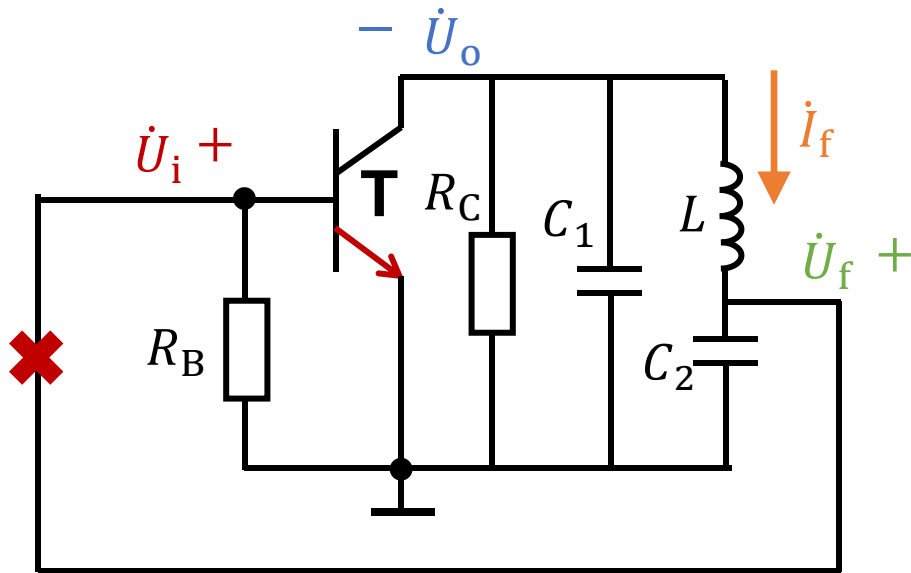
Positive or negative for AC circuit?



Positive or negative for AC circuit?

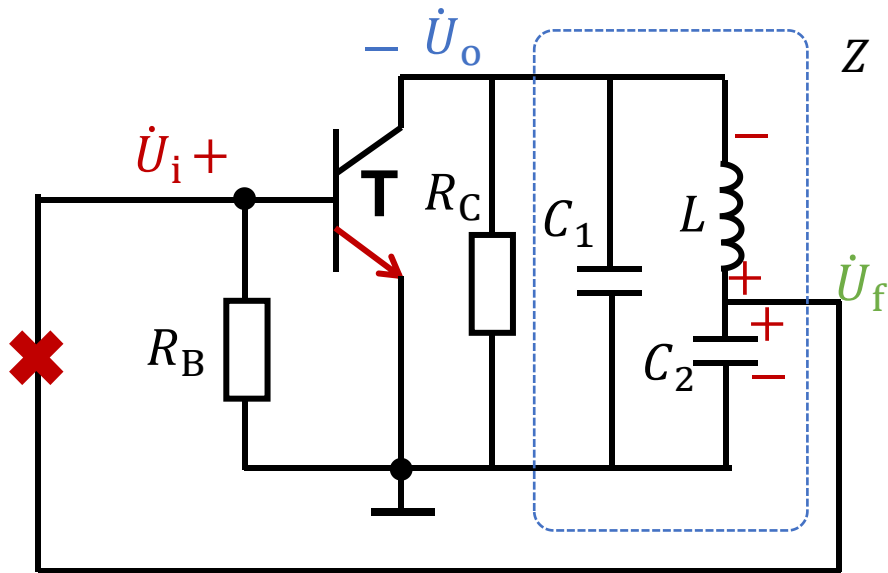


Positive or negative for AC circuit?

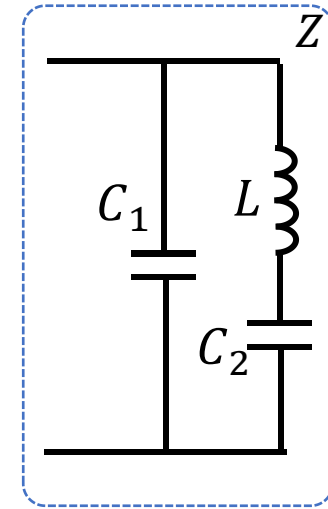


$\dot{U}_f$ and $\dot{U}_i$ have the same phase:
positive!

Resonant frequency? $\dot{A} = ?$ $\dot{F} = ?$



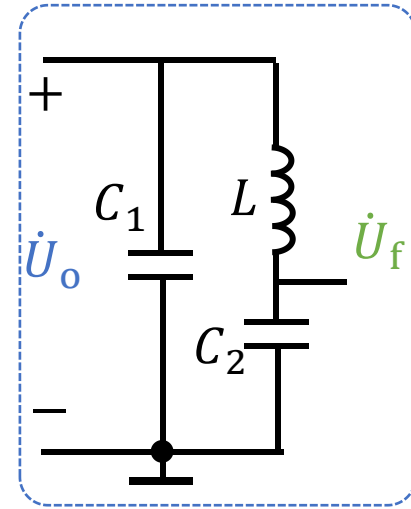
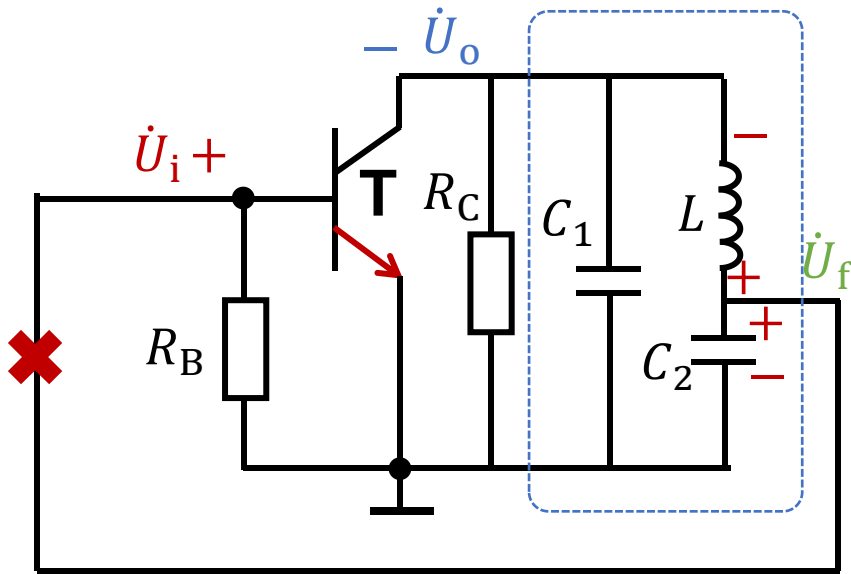
Voltage gain: $\dot{A} = -\frac{\beta(R_C || Z)}{r_{be}}$



Resonant frequency:

$$\left\{ \begin{array}{l} Z \rightarrow \infty \\ Y = 0 \end{array} \right. \leftarrow f_0 = \frac{1}{2\pi\sqrt{L(C_1 || C_2)}} \leftarrow \left\{ \begin{array}{l} Z = (j\omega L + \frac{1}{j\omega C_2}) // \frac{1}{j\omega C_1} \\ Y = \frac{1}{Z} = j(\frac{1}{\frac{1}{\omega C_2} - \omega L} + \omega C_1) \end{array} \right.$$

Positive feedback network:



$$\dot{F} = -\frac{C_1}{C_2}$$

$$f_0 = \frac{1}{2\pi\sqrt{L(C_1||C_2)}}$$

$$\dot{F} = \frac{\dot{U}_f}{\dot{U}_o} = \frac{\frac{1}{j\omega C_2}}{\frac{1}{j\omega C_2} + j\omega L} = \frac{1}{1 - \omega^2 C_2 L}$$

Start-oscillation condition:

(a) $\dot{A}\dot{F} > 1$

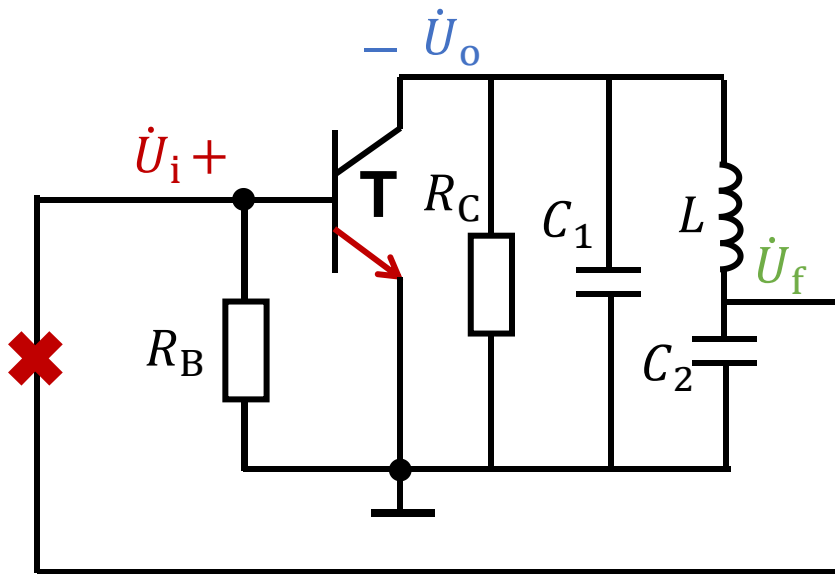
$$\left\{ \begin{array}{l} \dot{A} = -\frac{\beta(R_c \parallel Z)}{r_{be}} \\ \dot{F} = -\frac{C_1}{C_2} \end{array} \right.$$

(b) $\varphi = 2n\pi$

At resonant frequency: $f_0 = \frac{1}{2\pi\sqrt{L(C_1 \parallel C_2)}}$

$$\varphi_A = \pi, \quad \varphi_F = \pi.$$

$$\varphi = \varphi_A + \varphi_F = 2\pi$$



9.4 Quartz crystal oscillator 石英晶体振荡器

Quartz: SiO_2 crystal 二氧化硅晶体

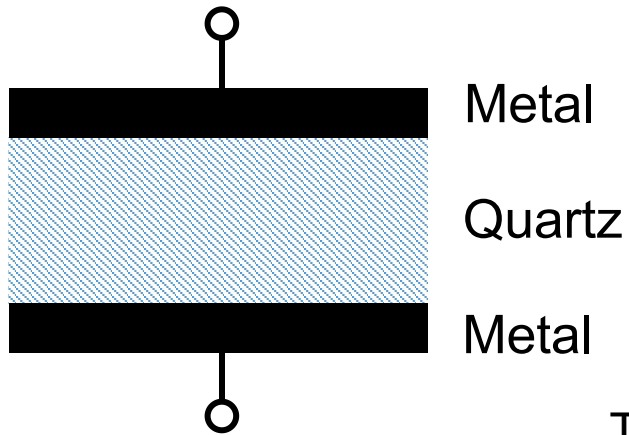
{ Crystalline anisotropy 晶体各向异性
Piezoelectric effect 压电效应

Piezoelectric effect:

Apply an electric field across quartz, there will be mechanical deformation.

Apply a mechanical force across quartz, there will be electric field.

Piezoelectric vibration 压电振荡



1) Apply a AC voltage across quartz;

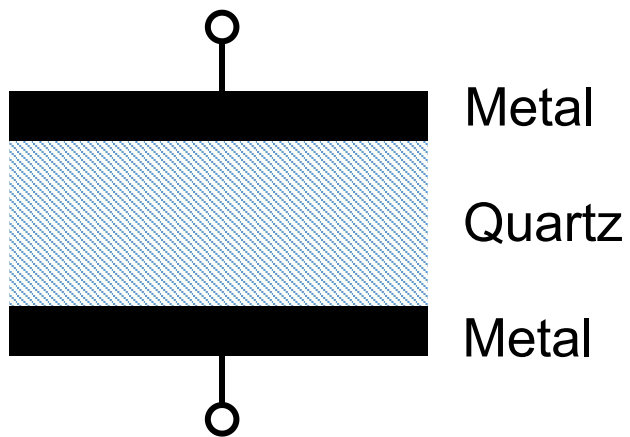
2) Mechanical vibration in quartz;

The amplitude of mechanical vibration is small

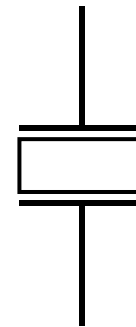
3) When AC voltage frequency equals to the intrinsic frequency of quartz.

4) The amplitude of vibration is very large, oscillation is resonant.

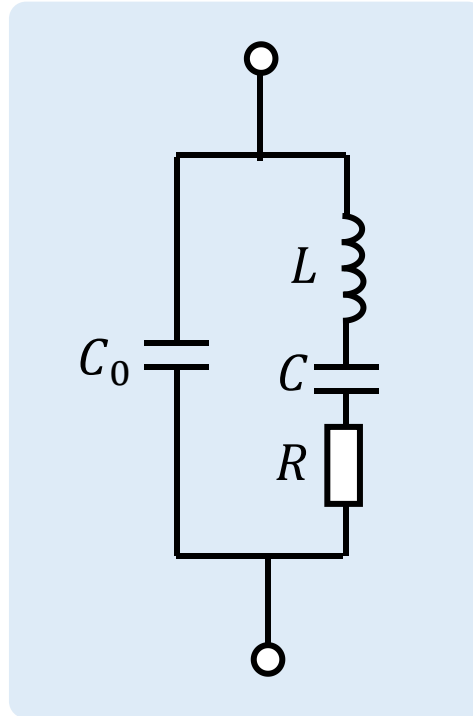
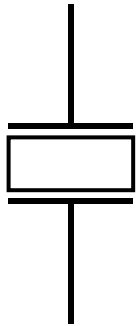
Intrinsic frequency: 固有频率, is also called resonant frequency 共振频率



Electronic circuit symbol



Equivalent circuit 等效电路



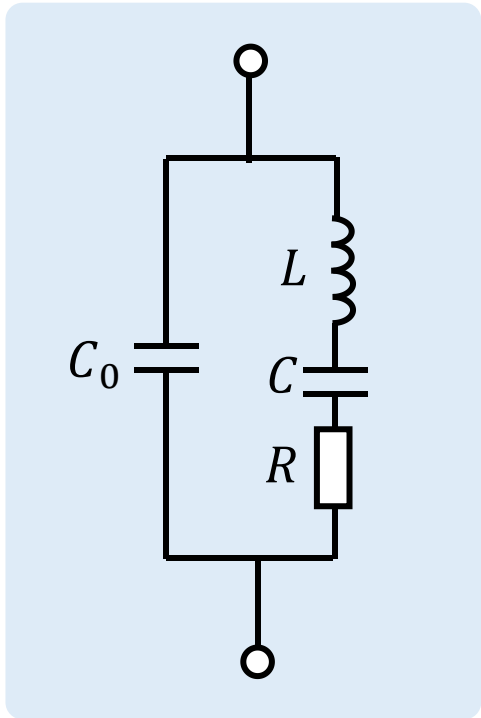
C_0 : Static capacitance

L : Inertia of mechanical vibration 机械振动的惯性

$C \ll C_0$: Elastic capacitance

R : Friction dissipation 摩擦损耗

Equivalent circuit 等效电路



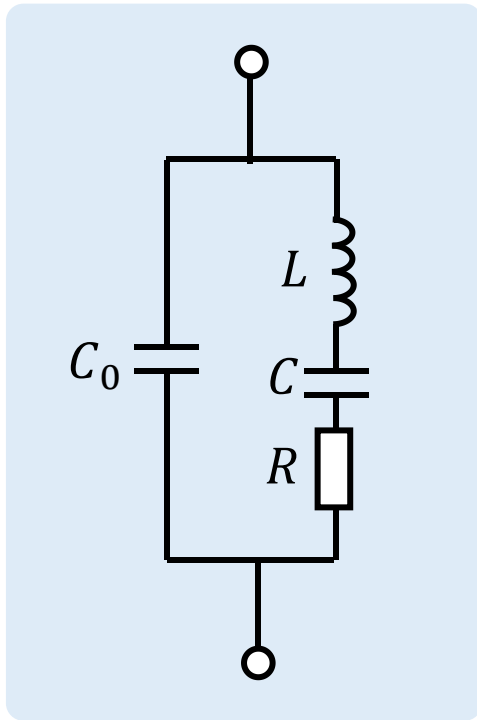
C_0 : Determine by area and thickness of quartz

L is large: 10 H ~ 10 mH

C is small: 0.01 ~ 0.1 pF

R is small: ~100 Ω

Equivalent circuit 等效电路



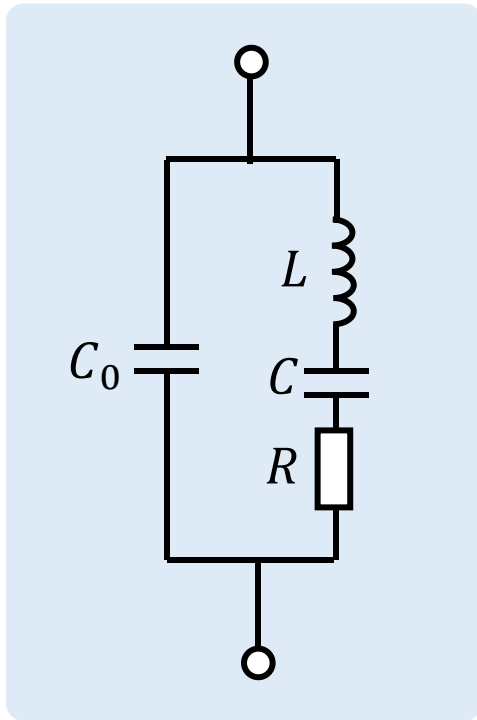
Quality factor Q is large: $10^4 \sim 10^6$

Frequency stability $\Delta f/f_0$ is very good:

- RC oscillator $\Delta f/f_0 : 10^{-2}$
- LC oscillator $\Delta f/f_0 : 10^{-3} \sim 10^{-4}$
- Quartz oscillator $\Delta f/f_0 : 10^{-9} \sim 10^{-11}$

Frequency response of impedance

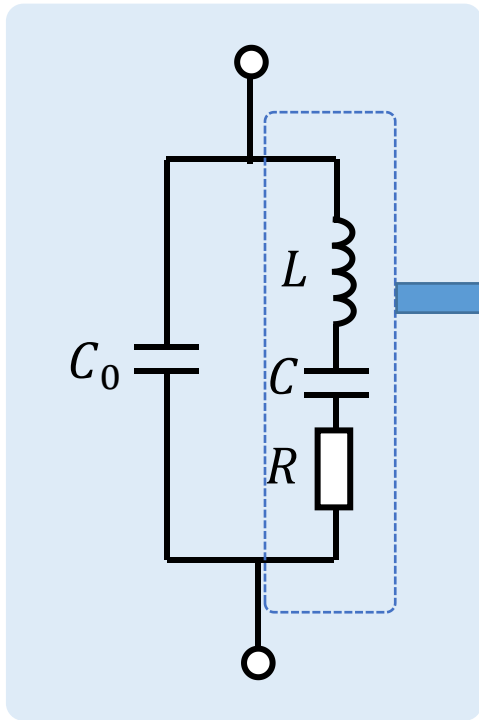
Ignore R: R=0



Impedance 阻抗 $\dot{Z}$ = reactance 电抗 $\dot{X}$

$$\begin{aligned}\dot{X} &= \frac{\frac{1}{j\omega C_0} (j\omega L + \frac{1}{j\omega C})}{\frac{1}{j\omega C_0} + j\omega L + \frac{1}{j\omega C}} \\ &= \frac{1 - \omega^2 LC}{j\omega C_0 (\frac{C}{C_0} + 1 - \omega^2 LC)}\end{aligned}$$

Frequency response of impedance



$$\dot{X} = \frac{1 - \omega^2 LC}{j\omega C_0 \left(\frac{C}{C_0} + 1 - \omega^2 LC \right)}$$

(a) When $1 - \omega^2 LC = 0$, $\dot{X} = 0$

Series resonance 串联谐振

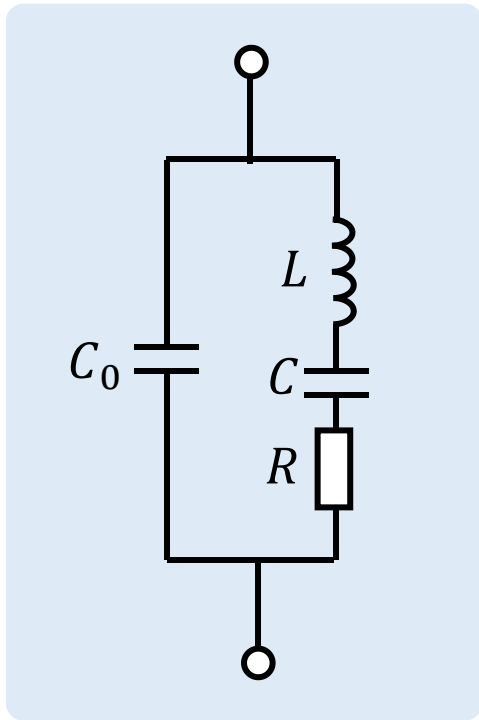
Series resonant frequency:

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

When $f < f_s$: circuit is capacitive 电路呈容性

When $f > f_s$: circuit is inductive 电路呈感性

Frequency response of impedance



$$\dot{X} = \frac{1 - \omega^2 LC}{j\omega C_0 \left(\frac{C}{C_0} + 1 - \omega^2 LC \right)}$$

(b) When $\frac{C}{C_0} + 1 - \omega^2 LC = 0$, $\dot{X} \rightarrow \infty$

Parallel resonance 并联谐振

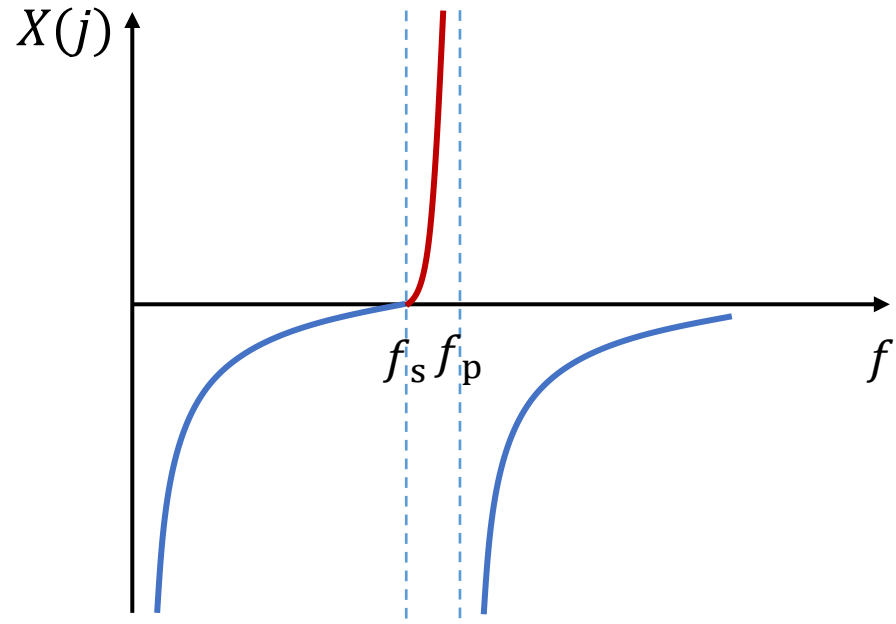
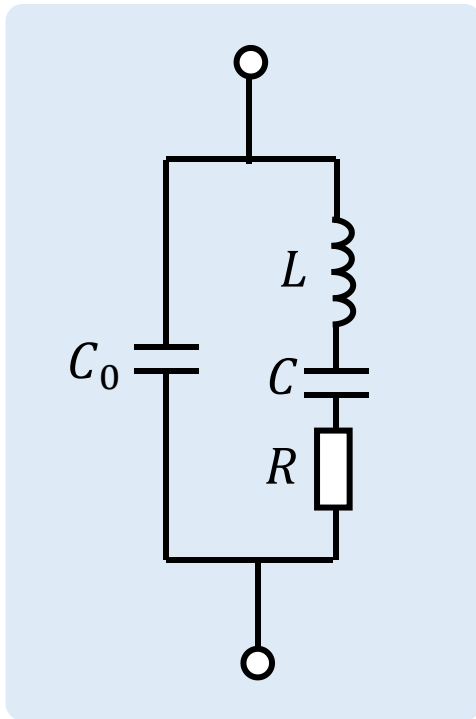
Parallel resonant frequency:

$$f_p = \frac{1}{2\pi\sqrt{LC||C_0}} > f_s$$

$$C \ll C_0: f_p \approx f_s$$

When $f > f_p$: circuit is capacitive 电路呈容性

Frequency response of impedance



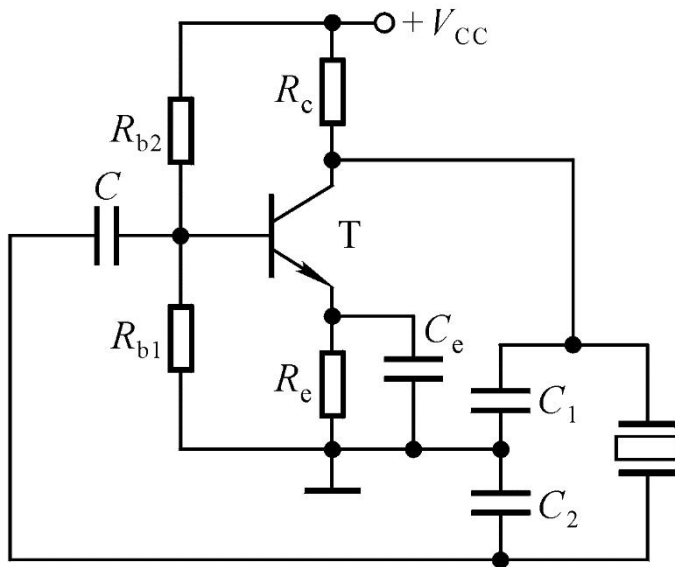
When $f < f_s$: circuit is capacitive

When $f_s < f < f_p$: circuit is inductive

When $f > f_p$: circuit is capacitive

Quartz oscillator circuit

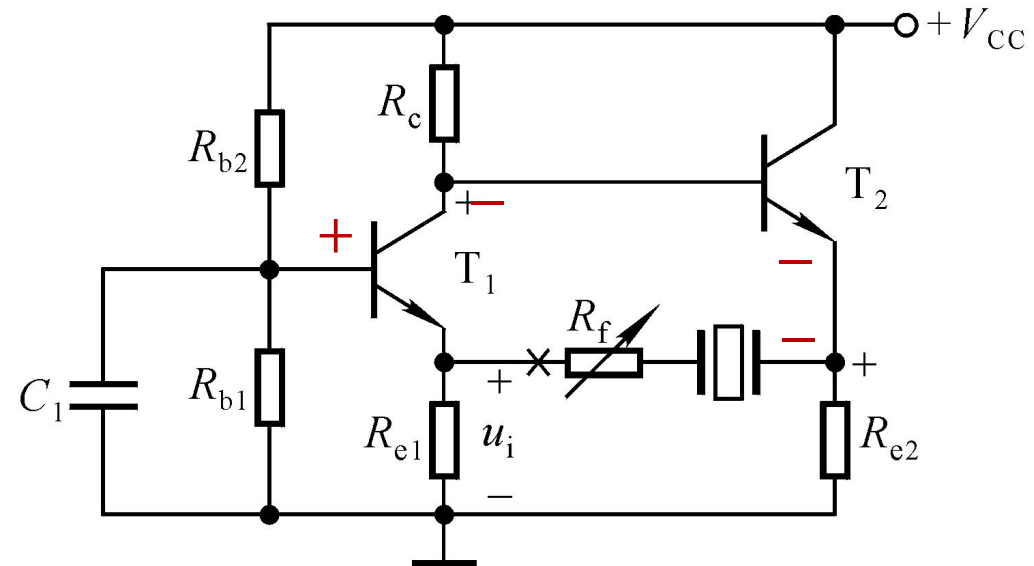
(a) Parallel type



Question 1: Quartz works at which region?

Question 2: Which type oscillator?

(b) Series type



Question 1: Quartz works at which region?

Homework 9-2: refer to blackboard and QQ